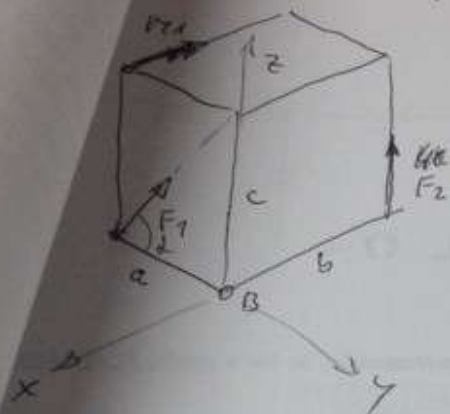


Statika 1



$$\begin{aligned} F_1 &= 8 \text{ kN} & a &= 2,4 \text{ m} \\ F_2 &= 4 \text{ kN} & b &= 3 \text{ m} \\ M_1 &= 5 \text{ kNm} & c &= 3,2 \text{ m} \end{aligned}$$

$$\sum F_{ix} = R_x = 0$$

$$\sum F_{iy} = R_y = F_1 \cdot \cos \alpha = 1,8 \text{ kN}$$

$$\sum F_{iz} = R_z = F_1 \cdot \sin \alpha + F_2 = 10,4 \text{ kN}$$

$$\sum M_{ix} = M_{xo} = -M_1 - F_1 \cdot \sin \alpha \cdot a$$

$$M_{xo} = -20,36 \text{ kNm}$$

$$\sum M_{iy} = M_{yo} = + F_2 \cdot b = 12 \text{ kNm}$$

$$\sum M_{iz} = M_{zo} = 0$$

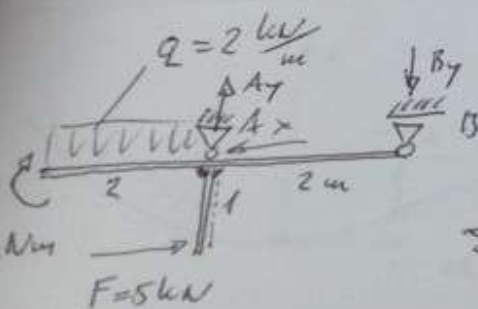
$$\vec{R} = (0, 1,8, 10,4) \text{ kN}$$

$$R = 11,45 \text{ kN}$$

$$\vec{M}_{oB} = (-20,36, 12, 0) \text{ kNm}$$

$$M_{oB} = 23,63 \text{ kNm}$$

2)



$$A_x = 5 \text{ kN}$$

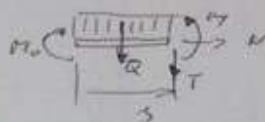
$$A_y - B_y - q \cdot 2 = 0 ; A_y = 5 \text{ kN}$$

$$\sum M_{ix} = 0 ; -M_o + q \cdot 2 \cdot 1 + F \cdot 1 - B_y \cdot 2 = 0$$

$$B_y = 1 \text{ kN}$$



levo polje:

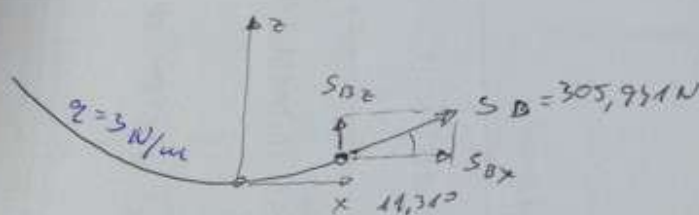
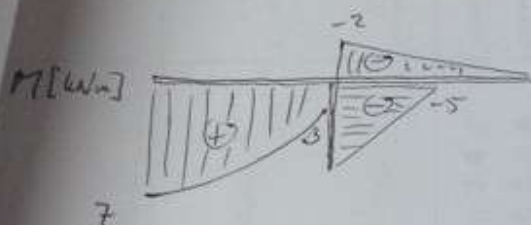


$$M - M_0 + 2 \cdot \frac{q \cdot 2}{2} = 0$$

$$M = M_0 - \frac{q}{2} \cdot 2^2$$

$$M(0) = M_0 = 2 \text{ kNm}$$

$$M(2) = 2 - \frac{q}{2} \cdot 2^2 = 0 \text{ kNm}$$



$$S_{Bx} = H = 300 \text{ N}$$

$$S_{Bz} = 60 \text{ N}$$

$$V_0 = x_0 \cdot q; \quad x_0 = \frac{60}{3} = 20 \text{ m}$$

$$x_A = l - x_0 = 60 - 20 = 40 \text{ m}$$

$$z_A = \frac{q \cdot x_A^2}{2H} = \frac{3 \cdot 40^2}{2 \cdot 300} = 8 \text{ m}$$

$$h = z_A - b = 6 \text{ m}$$

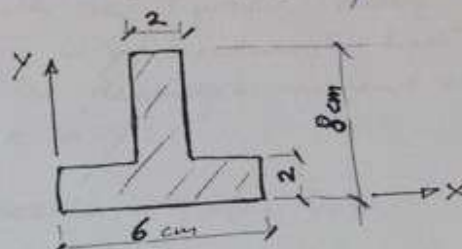
$$z_B = b = \frac{q \cdot x_0^2}{2H} = \frac{3 \cdot 20^2}{2 \cdot 300} = 2 \text{ m}$$

$$z_B = 2 \text{ m}$$

$$S_{max} = S_A = \sqrt{V_A^2 + H^2} = \underline{323,11 \text{ N}}$$

$$V_A = q \cdot x_A = 120 \text{ N}$$

- ① Izračunajte aksialne momente za T profil okrog težiščnih osi in okrog osi x in y ! Izračunajte tudi polarni vztrajnostni moment ter deviacijski moment za koordinatni sistem z osmi x in y !

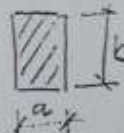
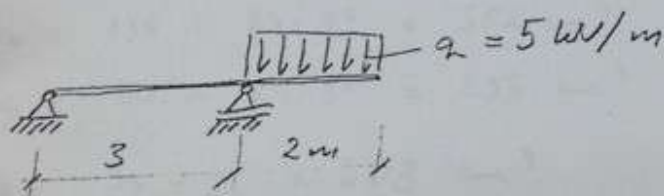


- ② Podan je deformacijski tenzor: $\epsilon_{ij} = \begin{pmatrix} 1 & -3 & 0 \\ -3 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \cdot 10^{-4}$

Izračunajte glavne normalne napetosti!

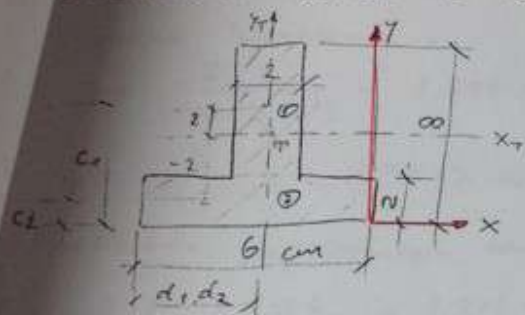
Modul elastičnosti: $E = 2,1 \cdot 10^5 \text{ N/mm}^2$, Poissonovo število $\nu = 0,3$.

- ③ Dimenzionirajte lesen nosilec na upogib, če je dopustna napetost $1,0 \text{ kN/cm}^2$! Prerez nosilca je pravokotnik, katerega stranici sta $a = 3x$ in $b = 5x$.



Višja, Visoka, Gozdarska (obrne x, y osi)

Izračunajte aksialne momente za T profil okrog žižnih osi in okrog osi x in y! Izračunajte tudi plani vztrajnostni moment, ter deviacijski moment za koordinatni sistem z osmi x in y!



težišče : $T = (x_T, y_T)$

$$x_T = -3 \text{ cm}$$

$$y_T = \frac{72}{24} = 3 \text{ cm}$$

i	A _i	y _i	A _i y _i
1	2.6	5	60
2	2.6	1	12
Σ	24		72

Težiščne osi :

$$J_{x_T} = \frac{2 \cdot 6^3}{12} + 2 \cdot 6 \cdot 2^2 + \frac{6 \cdot 2^3}{12} + 2 \cdot 6 \cdot 2^2 = 136 \text{ cm}^4$$

$$J_{y_T} = \frac{6 \cdot 2^3}{12} + \frac{2 \cdot 6^3}{12} = 40 \text{ cm}^4$$

Koord. sist. x, y

$$J_x = 136 + 24 \cdot 3^2 = 352 \text{ cm}^4$$

$$J_y = 40 + 24 \cdot 3^2 = 256 \text{ cm}^4$$

$$J_p = J_x + J_y = 608 \text{ cm}^4$$

$$J_{xy} = 12 \cdot (-3) \cdot 5 + 12 \cdot (-3) \cdot 1 = -216 \text{ cm}^4 \quad J_{xy} = \sum A_i \cdot c_i \cdot d_i$$

leska, gozdarska

Podan je deformacijski tenzor: $\epsilon_{ij} = \begin{pmatrix} 1 & -3 & 0 \\ -3 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \cdot 10^{-4}$
zračunajte glavne normalne napetosti in smeri glavnih osi.
Modul elastičnosti je $E = 2,1 \cdot 10^5 \text{ N/mm}^2$ in Poissonovo število 0,3.

Lamejeve konstante:

$$\lambda = \frac{E \nu}{(1-2\nu)(1+\nu)} = 1,211 \cdot 10^5 \text{ N/mm}^2$$

$$G = \frac{E}{2(1+\nu)} = 0,8077 \cdot 10^5 \text{ N/mm}^2$$

Volumenska dilatacija

$$\epsilon_v = \epsilon_x + \epsilon_y + \epsilon_z$$

$$\epsilon_v = 3 \cdot 10^{-4}$$

Koeficienti napetostnega tenzorja:

$$\sigma_x = \lambda \cdot \epsilon_v + 2G \epsilon_x = 1,211 \cdot 10^5 \cdot 3 \cdot 10^{-4} + 2 \cdot 0,8077 \cdot 10^5 \cdot 1 \cdot 10^{-4} = 52,484 \text{ N/mm}^2$$

$$\sigma_y = \lambda \cdot \epsilon_v + 2G \epsilon_y = 68,638 \text{ N/mm}^2$$

$$\sigma_z = \lambda \epsilon_v + 2G \epsilon_z = 36,33 \text{ N/mm}^2$$

$$\tau_{xy} = G \cdot 2 \cdot \epsilon_{xy} = 2 \cdot 0,8077 \cdot 10^5 \cdot (-3) \cdot 10^{-4} = -48,462 \text{ N/mm}^2$$

$$\sigma_{ij} = \begin{pmatrix} 52,484 & -48,462 & 0 \\ -48,462 & 68,638 & 0 \\ 0 & 0 & 36,33 \end{pmatrix} \text{ N/mm}^2$$

Normalne glavne napetosti:

$$\begin{vmatrix} (\sigma_x - \sigma) & \tau_{xy} & 0 \\ \tau_{yx} & (\sigma_y - \sigma) & 0 \\ 0 & 0 & (\sigma_z - \sigma) \end{vmatrix} = 0$$

$$\sigma_1 = 52,484 \text{ N/mm}^2$$

$$\sigma_2 = 36,33 \text{ N/mm}^2$$

$$(\sigma_z - \sigma) [(\sigma_x - \sigma)(\sigma_y - \sigma) - \tau_{xy}^2] = 0$$

$$\sigma_x \sigma_y - \sigma_y \sigma - \sigma_x \sigma + \sigma^2 - \tau_{xy}^2 = 0$$

$$\sigma^2 - \sigma(52,484 + 68,638) - 48,462^2 + 52,484 \cdot 68,638 = 0$$

$$\sigma^2 - 121,12 \sigma + 1253,7 = 0$$

$$\sigma_{1,2} = \frac{121,12 \pm \sqrt{121,12^2 - 4 \cdot 1 \cdot 1253,7}}{2}$$

$$\sigma_1 = 109,7 \text{ N/mm}^2$$

$$\sigma_3 = 11,43 \text{ N/mm}^2$$

Dimenzionirajte lesen nosilec na upogib, če je dopustna napetost $1,0 \text{ kN/cm}^2$! Prerez nosilca je pravokotnik, katerega stranici sta $a = 3x$ in $b = 5x$.



$$q = 5 \text{ kN/m}$$



$$M_{\max} = q \cdot \frac{2^2}{2} = 5 \cdot 2 = 10 \text{ kNm}$$

$$J_x = \frac{3x \cdot (5x)^3}{12} = 31,25 x^4$$

$$\sigma_{\text{dop}} = 1,0 \text{ kN/cm}^2$$

$$\sigma = \frac{M \cdot c}{J} \Rightarrow$$

$$1 = \frac{10 \cdot 1000 \cdot 2,5x}{31,25 x^4}$$

$$x = \sqrt[3]{\frac{2500}{31,25}} = 4,31 \text{ cm}$$

nosilec ima izmere: $13 \times 22 \text{ cm}$

Trdnost OGR - VSP

23.1.2006

1) Podan je deformacijski tenzor: $\epsilon_{ij} = \begin{pmatrix} 1 & -3 & 0 \\ -3 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \cdot 10^{-4}$

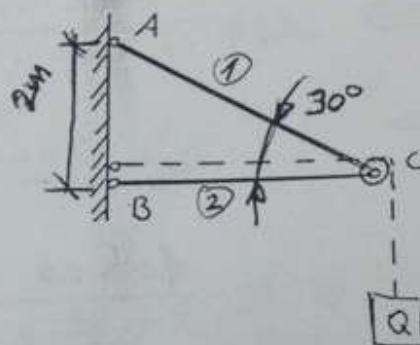
Izračunajte glavne normalne napetosti!

Modul elastičnosti: $E = 2,1 \cdot 10^5 \text{ N/mm}^2$, Poissonovo število $\nu = 0,3$.

2) Vračunajte napetosti v palicah in premiku točke C!

$$A_1 = A_2 = 6 \text{ cm}^2$$

$$Q = 30 \text{ kN}, E = 210000 \text{ MPa}$$

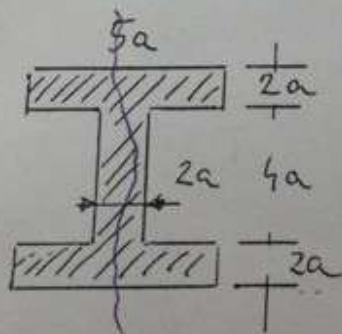


3) Izračunajte velikost največje upogibne napetosti v nosilec AB! Označite mesto na sliki!

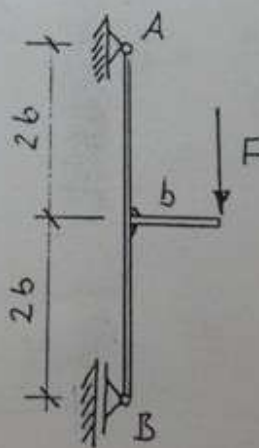
$$a = 2 \text{ m}, b = 0,5 \text{ m}$$

$$F = 400 \text{ N}$$

prerez:



I - pogled na nosilec AB z vrha



$$\nu = 3 \text{ (uklonška varnost)}$$

Izračun dovoljen mo Eulerje, Tetmajerje

$$1) \quad \varepsilon = \begin{pmatrix} 1 & -3 & 0 \\ -3 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \cdot 10^{-4}$$

$$E = 2,1 \cdot 10^5 \text{ MPa}$$

$$\nu = 0,3$$

$$\lambda = \frac{E \cdot \nu}{(1+\nu)(1+2\nu)} = 1,211 \cdot 10^5 \text{ MPa}$$

$$G = \frac{E}{2(1+\nu)} = 0,8077 \cdot 10^5 \text{ MPa}$$

$$\varepsilon_v = 3 \cdot 10^{-4}$$

$$\sigma_x = \lambda \cdot \varepsilon_v + 2 G \cdot \varepsilon_x = 52,48 \text{ MPa}$$

$$\sigma_y$$

$$\sigma_z$$

$$\tau_{xy} = 2 \cdot G \cdot \varepsilon_{xy} = -48,462 \text{ MPa}$$

$$\sigma_{ij} = \begin{pmatrix} 52,48 & -48,462 & 0 \\ -48,462 & 68,638 & 0 \\ 0 & 0 & 36,33 \end{pmatrix} \text{ MPa}$$

$$\sigma_1 = 109,7 \text{ MPa}$$

$$\sigma_2 = 36,33 \text{ MPa}$$

$$\sigma_3 = 11,43 \text{ MPa}$$

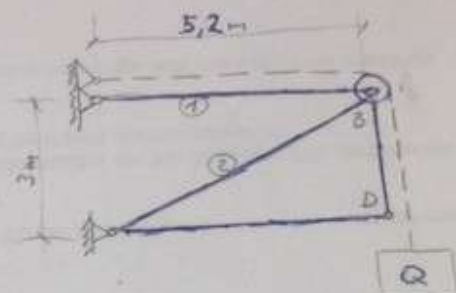
3. kvatrolerij iz mehanike -
lesarji, univerzitetni študij
16. 4. 2022

- 1) Izračunajte minimalne
velikosti presezov palic 1 in 2,
tako, da ne bo presežena
dopustna napetost $\sigma_{dop} = \pm 70 \text{ MPa}$!

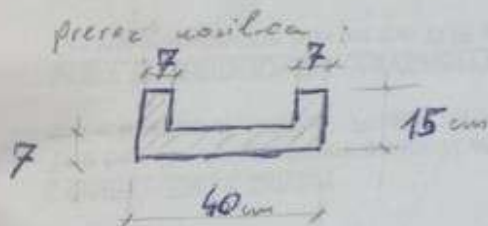
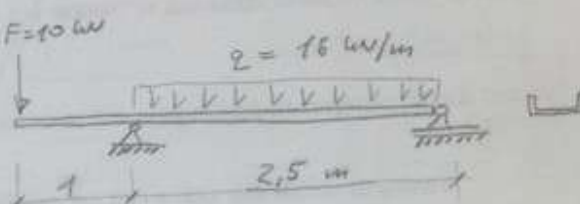
Izračunajte premike točke D (x-y)!

$E = 0,7 \cdot 10^5 \text{ MPa}$, $Q = 60 \text{ kN}$

Trenja ni! Ne upoštevajte uklona! Dimenzije uklona v točki B
zanemarite!



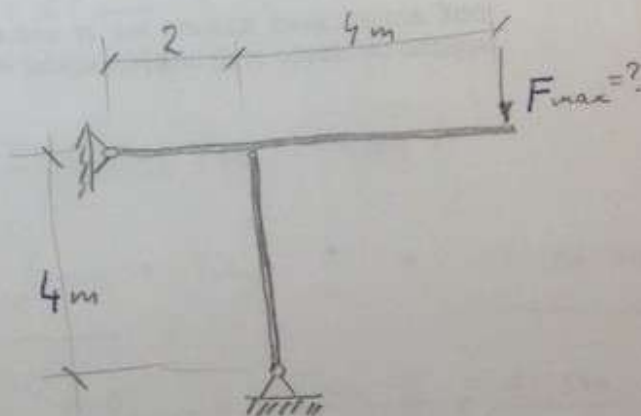
- 2) Izračunajte velikosti in
mesta (kde) ekstremnih
napetosti! Narišite
diagram upogibnih momentov!
prerez nosilca:



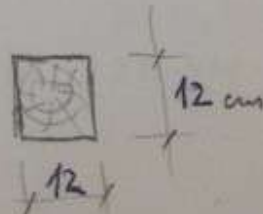
- 3) Izračunajte maksimalno silo,
da se stebel ne ukloni!
Stebel je lesnik (iglavci), kvadratnega
prereza. $E_{II} = 1000 \text{ kN/cm}^2$.

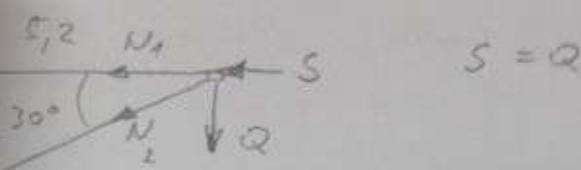
$\sigma_{dop} = 1,0 \text{ kN/cm}^2$

$\nu = 3$ (uklonška varnost)



Izračun dovoljen po Eulerju, Tetmajerju
ali "w" postopku





$$-N_1 - S - N_2 \cdot \cos 30^\circ = 0$$

$$-N_2 \cdot \sin 30^\circ - Q = 0$$

$$N_2 = -\frac{Q}{\sin 30^\circ} = -120 \text{ kN}$$

$$N_1 = -S - N_2 \cdot \cos 30^\circ$$

$$N_1 = -60 + 120 \cdot \cos 30^\circ = 43,92 \text{ kN}$$

palica: $\sigma = \frac{N_1}{A_1}$; $A_1 = \frac{N_1}{\sigma_{\text{adap}}} = \frac{43,92 \text{ kN}}{7 \text{ MPa}}$

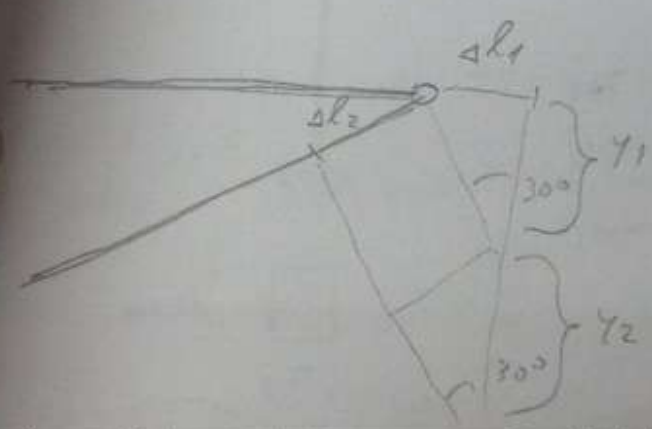
$$A_1 = 6,275 \text{ cm}^2$$

$$\sigma = E \cdot \epsilon; \frac{\Delta l}{l} = \frac{\sigma}{E}; \Delta l_1 = l_1 \cdot \frac{\sigma}{E}$$

$$\Delta l_1 = 5200 \cdot \frac{70}{97.000} = 5,2 \text{ mm}$$

2 palica: $A_2 = \frac{N_2}{\sigma_{\text{adap}}} = \frac{-120}{-7} = 17,14 \text{ cm}^2$

$$\Delta l_2 = 6003,33 \cdot \frac{-70}{70000} = -6,003 \text{ mm}$$



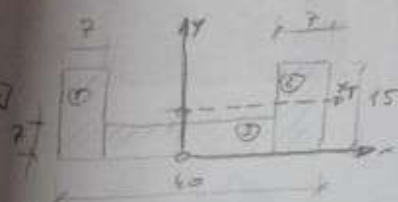
$$\tan 30^\circ = \frac{\Delta l_1}{l_1}; l_1 = \frac{5,2}{\tan 30^\circ} = 9 \text{ mm}$$

$$\sin 30^\circ = \frac{\Delta l_2}{l_2}; l_2 = \frac{6,003}{\sin 30^\circ} = 12 \text{ mm}$$

$$y_B = 9 + 12 = 21 \text{ mm}$$

Euler: $F_{ik} = \left(\frac{\pi}{\beta l}\right)^2 E J_{\min} = \frac{\pi^2 \cdot 10000 \cdot 1728}{600^2} = 106,59 \text{ kN}$

glej: uvelogo izpit iz reakt. - lesarji - 18. 4. 96



i	A _i	y _i	A _i · y _i
1	105	7,5	787,5
2	105	7,5	787,5
3	182	3,5	637
Σ	392		2212

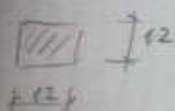
$$y_T = 5,6428 \text{ cm}$$

$$J_{xT} = 2 \cdot \left[\frac{7,5 \cdot 15^3}{12} + 105 \cdot (7,5 - 5,64)^2 \right] + \frac{26 \cdot 7^3}{12} + 182 \cdot (5,64 - 3,5)^2$$

$$J_{xT} = 6240,6 \text{ cm}^4$$

upogibne napetosti: $\sigma_a = \frac{M \cdot y_a}{J} = \frac{1000 \cdot (15 - 5,6428)}{6240,6} = 1,4934 \text{ kN/cm}^2$

$$\sigma_t = \frac{1000 \cdot 5,6428}{6240,6} = 0,9042 \text{ kN/cm}^2$$



$$A = 144 \text{ cm}^2$$

$$J = 1728 \text{ cm}^4$$

$$i = \sqrt{\frac{J}{A}} = 3,464 \text{ cm}$$

$$\beta = 1, \lambda = \frac{400}{3,464} = 115,47 - \text{euler ali } w$$

$$\text{Euler:)} F_k = \left(\frac{\pi}{\beta l} \right)^2 E J_{\min} = \frac{\pi^2 \cdot 1000 \cdot 1728}{400^2} = 106,59 \text{ kN}$$

$$F_{dop} = F_k / 3 = 35,53 \text{ kN} ; F_{\max} = F_{dop} \cdot \frac{2}{6} = 11,84 \text{ kN}$$

$$w) \lambda = 115,47 \Rightarrow w = 4,00743$$

$$F_{dop} = \frac{F_k \cdot 1}{w} = \frac{1 \cdot 144}{4,00743}$$

$$F_{dop} = 35,933 \text{ kN}$$

$$F_{\max} = 11,377 \text{ kN}$$

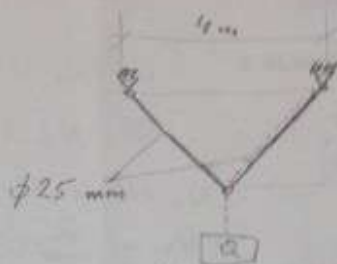
4. popravni kolokvij iz tehnike mehanike
26.6.22, 9.9.22

12. točka

- 1) Izračunajte pomak
točke C!

$$Q = 100 \text{ kN}$$

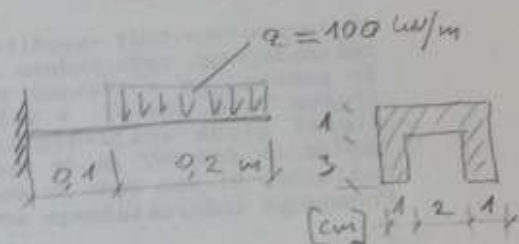
$$E = 2,1 \cdot 10^5 \text{ MPa}$$



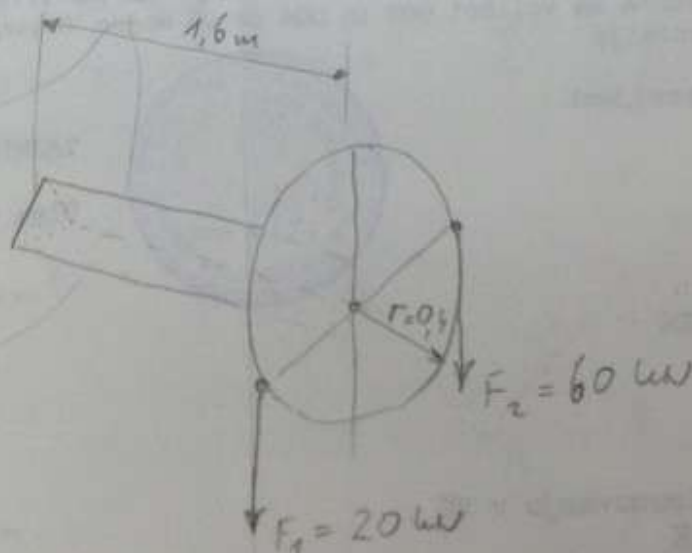
popravi na 2m

točka geometrija

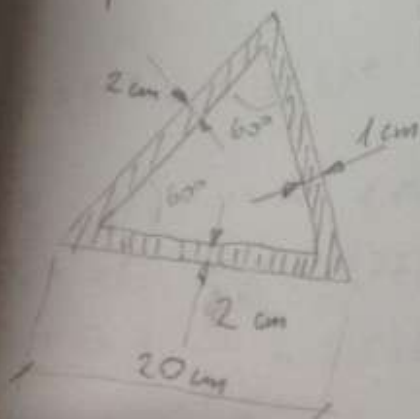
- 2) Izračunajte veličnost najveće
strižne napetosti v nosilcu!



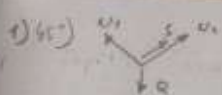
- 3) Izračunajte veličnost najveće
torzijske napetosti v
prerezu!



prerez nosilca



geometrija



$$\begin{aligned} -N_1 \cos 33^\circ + N_2 \cos 33^\circ + S \cos 45^\circ &= 0 \\ N_1 \sin 33^\circ + N_2 \sin 33^\circ + S \sin 45^\circ &= Q \end{aligned}$$



$$\begin{aligned} -N_1 \cos \alpha + S \cos \alpha + N_2 \cos \alpha &= 0 \\ N_1 \sin \alpha + S \sin \alpha + N_2 \sin \alpha - Q &= 0 \\ -N_1 + S + N_2 &= 0 \\ N_1 + S + N_2 &= \frac{Q}{\sin \alpha} = 120,16 \end{aligned}$$

$$2S + 2N_2 = 120,16$$

$$2N_2 = -79,81$$

$$N_2 = -39,91 \text{ kN}$$

$$N_1 = 60,09 \text{ kN}$$

povz. točke C ; $\frac{F}{l} = E \cdot \frac{\Delta l}{l}$

$$C_x = \frac{D_x + E_x}{2} = 3,152 \text{ mm}$$

$$C_y = \frac{E_y - D_y}{2} \cdot \tan \beta = -0,427 \text{ mm}$$

$$\Delta l_1 = \frac{N_1}{A_1} \cdot \frac{l_1}{E} = \frac{60090 \cdot 3605}{490 \cdot 211 \cdot 10^3}$$

$$\Delta l_1 = 2,1 \text{ mm}$$

$$\Delta l_2 = \frac{-39910 \cdot 3605}{490 \cdot 211 \cdot 10^3} = -1,395 \text{ mm}$$

$$l_1 = l_2 = 3605,5 \text{ mm}$$

$$A_1 = A_2 = \frac{\pi \cdot 25^2}{4} = 490,87 \text{ mm}^2$$

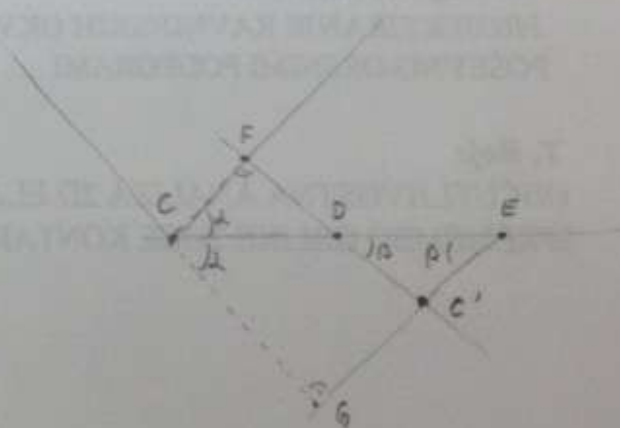
$$\rho = 90^\circ - \alpha = 33,69^\circ$$

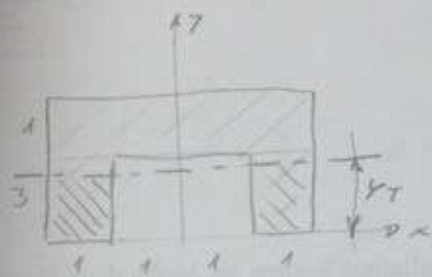
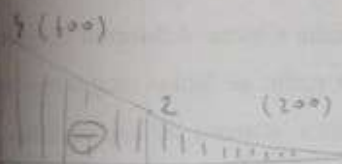
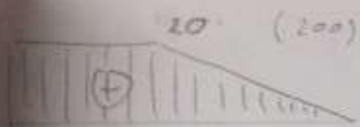
$$F_y = \Delta l_2 \cdot \sin \alpha = 1,161 \text{ mm}$$

$$F_x = \Delta l_2 \cdot \cos \alpha = 0,777 \text{ mm}$$

$$D_x = F_x + \frac{F_y}{\tan \beta} = 2,516 \text{ mm}$$

$$E_x = G_x + \frac{G_y}{\tan \beta} = \Delta l_1 (\cos \alpha + \frac{\sin \alpha}{\tan \beta}) = 3,788 \text{ mm}$$





i	A _i	y _i	A _i · y _i
1	4	3,5	14
2	6	1,5	9
Σ	10		23

$$y_T = \frac{23}{10} = 2,3 \text{ cm}$$

$$\tau = \frac{Q \cdot S_T}{b \cdot J_x}$$

$$J_{x_T} = \frac{1 \cdot 4}{12} + 4 \cdot 1,2^2 + 2 \left(\frac{1 \cdot 3^3}{12} + 3 \cdot 0,8^2 \right)$$

$$J_x = 14,43 \text{ cm}^4$$

$$\tau = \frac{20 \cdot 5,29}{2 \cdot 14,43}$$

$$S_{y_T} = \frac{y_T}{2} \cdot 2 \cdot y_T \cdot 1 = 5,29 \text{ cm}^3$$

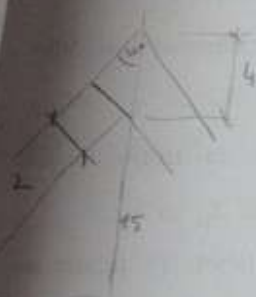
$$\tau = 3,666 \text{ W/cm}^2$$

$$M_t = F_1 \cdot 0,4 - F_2 \cdot 0,1 = 16 \text{ kNm}$$

$$W_t = (A_{zun} + A_{notr}) \cdot t_{min}$$



$$A_{zun} = 20 \cdot 12,32 / 2 = 123,2 \text{ cm}^2$$



$$\sin 30^\circ = \frac{3}{x} ; x = \frac{3}{\sin 30^\circ} = 6 \text{ cm}$$

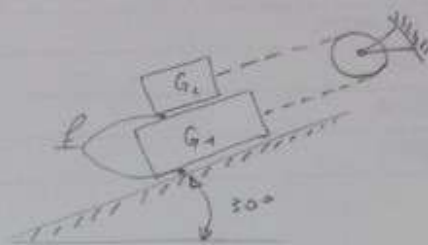
$$A_{notr} = 14,22 \cdot 12,32 / 2 = 87,6 \text{ cm}^2$$

$$W_t = 260,6 \text{ cm}^3$$

$$\sigma = \frac{M_t}{W_t} = \frac{1600}{260,6} = 6,139 \text{ kN/cm}^2$$

Izpit iz Tehniške mehanike - lesarje
4.6.98

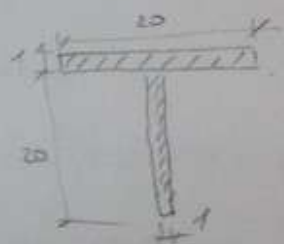
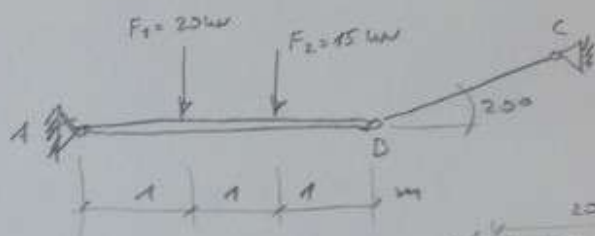
- 1) Določite minimalno težo telesa G_2 , da bo sistem se miroval! Trenje na kolenu zanemarite!
 $G_1 = 10 \text{ kN}$, $f = 0,1$



(glej izpit Stat in Kinet, p. 6.94)

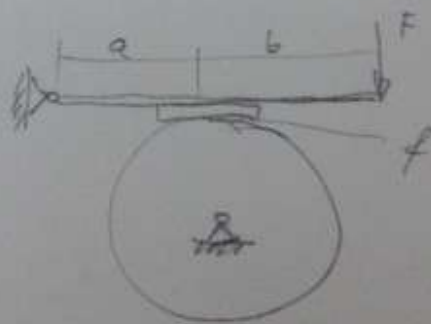
- 2) Za jeklen nosilec AB izračunajte lego in velikost največje in najmanjše normalne napetosti ter lastno frekvenco upogibne nihanj. Poves nosilca na mestu pod silo F_i je:

$$f_i = \frac{F_i}{3EI} \cdot \frac{a_i^2 \cdot b_i^2}{L}$$



3)

glej 4 kolokvij iz
Dinamike in Hidromehanike
500 kj.
(ni datuma)



$$F_1 = 20 \text{ kN}$$

$$F_2 = 15 \text{ kN}$$

$$\sigma = \frac{F}{3EI} \frac{a^2 \cdot b^2}{L}$$

$$E = 2,1 \cdot 10^5 \text{ kN/m}^2$$

Reakcije:

$$A_x + B \cos 20^\circ$$

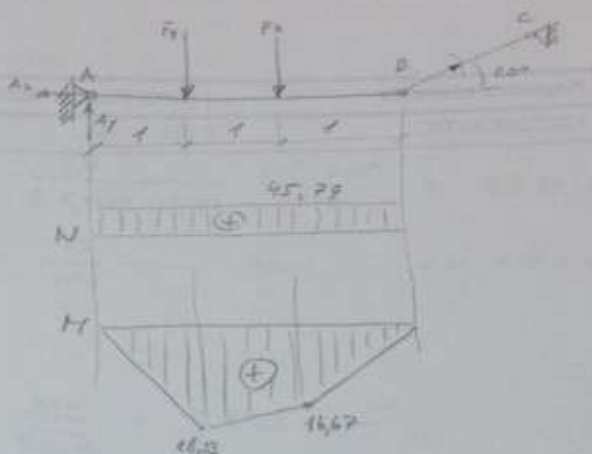
$$A_y + B \sin 20^\circ = F_1 + F_2$$

$$F_1 \cdot 1 = F_2 \cdot 2 + B \sin 20^\circ \quad \sum M = 0$$

$$B = 18,73 \text{ kN}$$

$$A_x = 45,79 \text{ kN}$$

$$A_y = 18,33 \text{ kN}$$



i	A _i	y _i	A _i · y _i
1	20	20,5	410
2	20	10	200
Σ	40		610

$$A = 40 \text{ cm}^2$$

$$y_T = \frac{610}{40} = 15,25 \text{ cm}$$

$$J_{xT} = \frac{20 \cdot 12^3}{12} + 20 \cdot (20,5 - 15,25)^2 + \frac{4 \cdot 20^3}{12} + 20 \cdot (15,25 - 10)^2$$

$$J_{xT} = 1770,8 \text{ cm}^4$$

Napetosti (vzpored + udiagonal)

$$\sigma_{2y} = + \frac{45,79}{40} - \frac{1833 \cdot (21 - 18,33)}{1770,8}$$

$$\sigma_{2y} = 1,144 - 5,95 = -4,8 \text{ kN/cm}^2$$

$$\sigma_{3y} = + \frac{45,79}{40} + \frac{1833 \cdot 15,25}{1770,8}$$

$$\sigma_{3y} = 1,144 + 15,785 = +16,93 \text{ kN/cm}^2$$

$$f_{st} = \frac{\sum f_i^2 F_i}{\sum f_i F_i} \quad - \text{Kull}$$

$$f_{st} = \sum f_i \quad - \text{Dumbarley}$$

$$f_1 = \frac{F_1}{3ES} \cdot \frac{a_1^2 \cdot b_1^2}{L} = \frac{20}{3 \cdot 2,1 \cdot 10^3 \cdot 11798} \cdot \frac{100^3 \cdot 200^3}{300} = 0,239 \text{ cm}$$

$$f_2 = \frac{F_2}{\dots} = \frac{15}{1260} \cdot 1260 = 0,1992 \text{ cm}$$

$$f_{st} = \sum f_i = 0,4383 \text{ cm}$$

$$v_0 = \frac{300}{\sqrt{f_{st}}} = \frac{300}{\sqrt{0,4383}} = 463,8 \text{ min}^{-1}$$

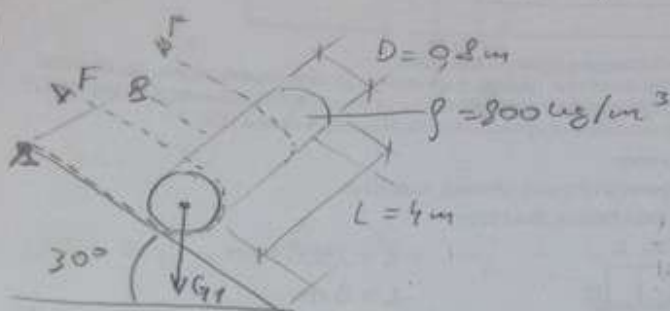
$$\omega_0 = \frac{\pi v_0}{30} = 48,57 \text{ rad}^{-1}$$

$$T_0 = \frac{2\pi}{\omega_0} = 0,1293 \text{ s}$$

$$v_s = 7,73 \text{ Hz}$$

Gozdarstvo, izpit 11. 4. 2013

1)



delo

$$V = \frac{0.8^2 \cdot \pi}{4} \cdot 4$$

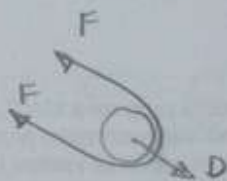
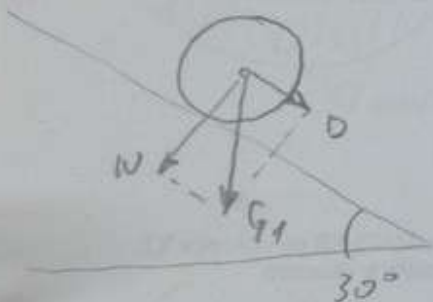
$$V = 2.011 \text{ m}^3$$

$$G = \rho \cdot V \cdot g = 800 \cdot 2.011 \cdot 10$$

$$G = 16085 \text{ N}$$

na eno vrvo pride polovična obtežba: $G_1 = 8042 \text{ N}$

$$D = G_1 \cdot \sin 30^\circ = 4021 \text{ N}$$



$$F = \frac{D}{2} = 2011 \text{ N}$$

Fpretirna vsaj 6x večja od $F: F_{\text{pretir}} = 12064 \text{ N}$

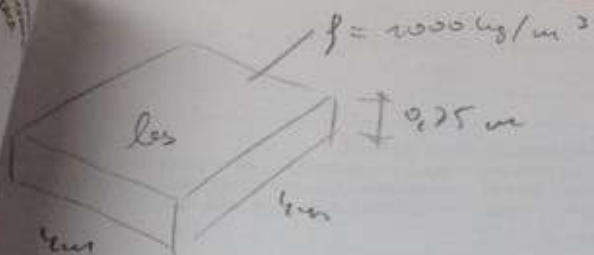
$$F_{\text{pretir}} = 12.1 \text{ kN}$$

programa MATLAB.
na nosilcu, na katerem je makromodel
y urejeni obliki z uporabo MATLAB-ovega ukaznega jezika.
porabo MATLAB-ovega ukaznega jezika.

Priimek_Ime1Et.m
Et.m

$q = 7500 \text{ N/m}$
 $L = 5 \text{ m}$
 $E = 2 \cdot 10^5 \text{ MPa}$
 $J = 7 \cdot 10^2 \text{ cm}^4$

$+2\left(\frac{x}{L}\right)^3$



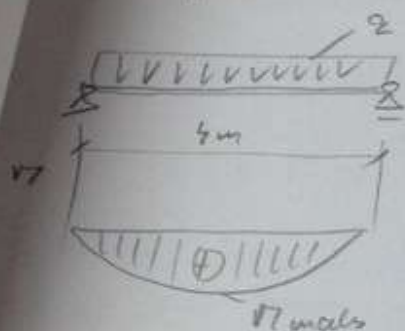
$$V = 0.75 \cdot 4 \cdot 4 = 12 \text{ m}^3$$

$$m = \rho \cdot V = 12000 \text{ kg}$$

$$G = g \cdot m = 120000 \text{ N}$$

$$\text{Na en nosilec } G_1 = \frac{G}{2} = 60000 \text{ N}$$

$$q = \frac{G_1}{L} = 15000 \text{ N/m}$$



$$M_{\text{max}} = \frac{q L^2}{8} = \frac{15000 \text{ N} \cdot 4^2 \text{ m}^2}{8}$$

$$M_{\text{max}} = 30000 \text{ Nm} = 3000 \text{ kNm}$$

$$\sigma_{\text{dop}} = 160 \text{ N/mm}^2 = 16 \text{ kN/cm}^2$$

$$\sigma = \frac{M}{W}; \quad W_{\text{min}} = \frac{M}{\sigma_{\text{dop}}}$$

$$W_{\text{min}} = \frac{3000 \text{ kNm cm}^2}{16 \text{ kN/cm}^2}$$

$$W_{\text{min}} = 187.5 \text{ cm}^3$$

Izberem NPI 200

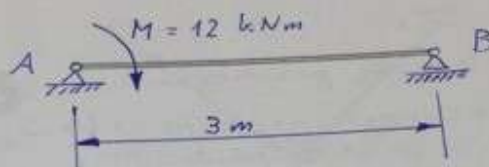
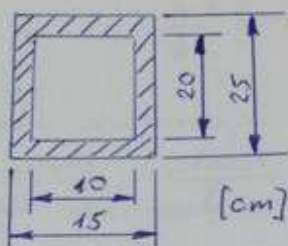
$$W_{\text{max}} = 214 \text{ cm}^3$$

2. Kolokvij iz Tehničke mehanike
27. 5. 1999. - godištno

- 1) Določite zaslek nosilca
v točki A!

$$E = 1000 \text{ kN/cm}^2$$

Prerez nosilca je
škatlast

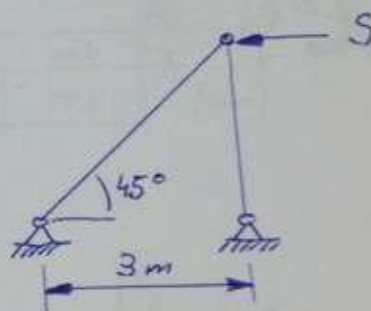


- 2) Izračunajte dopustno
silo S tako, da se
palica ① še ne ukloni!

$$E = 21000 \text{ kN/cm}^2$$

$$\nu = 3$$

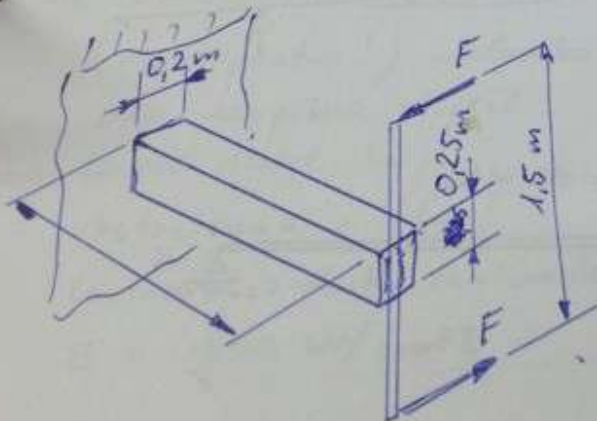
$$\tau_{\text{dop}} = 16 \text{ kN/cm}^2$$



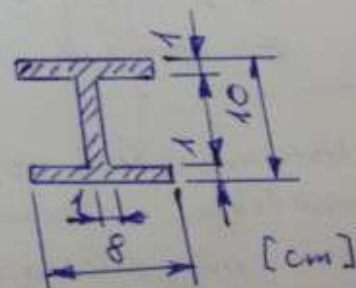
- 3) Kolikšen torzijski moment M_t
zdrži lesen ~~tram~~ tram?

$$\tau_{\text{dop}} = 6 \text{ MPa}$$

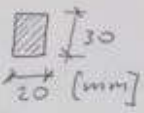
Izračunajte velikost sile F

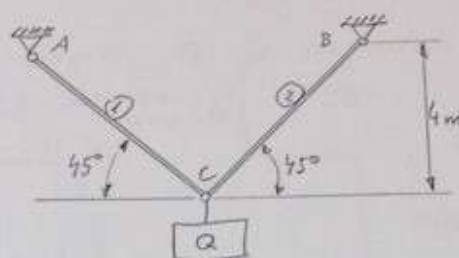


Prerez palice:



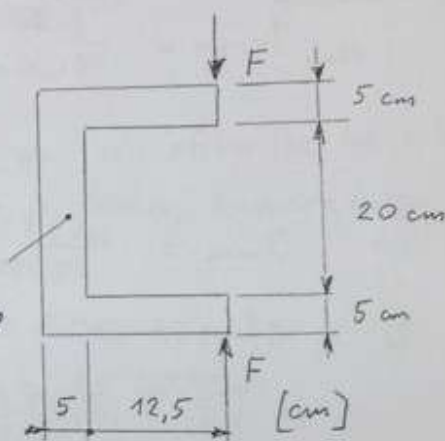
2. kolokvij iz mehanike - gozdarstvo
31.5.2000

- 1) Izračunajte pravo točke C
in napetosti v petlah 1 in 2!
 $G = 50 \text{ kN}$, $E = 2,1 \cdot 10^5 \text{ MPa}$
Prerez palic 1 in 2:  20 [mm]

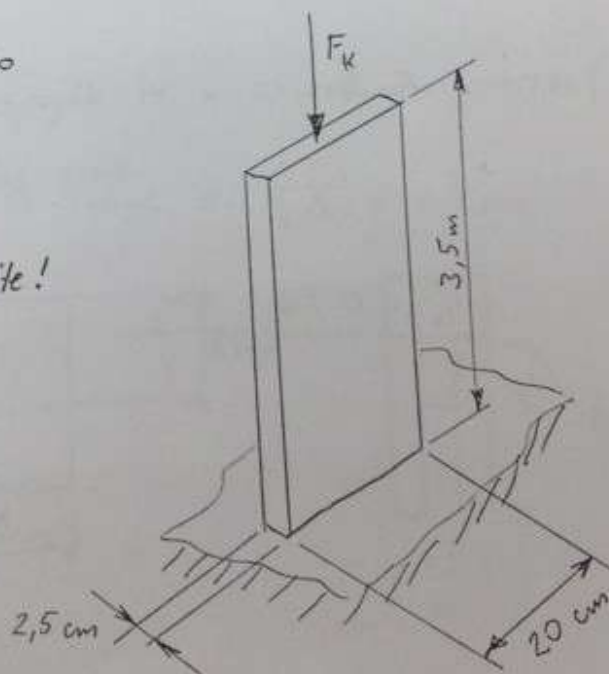


- 2) Za narisano konstrukcijo
določite mesto, velikost
in predznak ekstremalnih
upogibnih napetosti!
 $F = 400 \text{ N}$

konstantna
debelina 1,5 cm

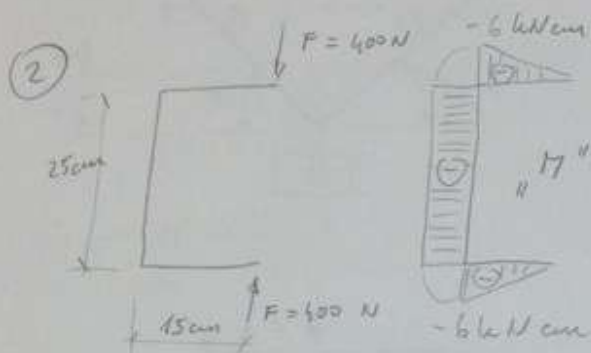


- 3) Izračunajte kritično
(ne dopustno!) uklonsko
silo navpično stojče
lesene deske, ki je
vbetonirana v tla!
Lastno težo deske zanemarite!
 $E = 1000 \text{ kN/cm}^2$



$$\textcircled{1} \quad \sigma_1 = \sigma_2 = \frac{F}{A} = \frac{35355}{600} = 58,925 \text{ N/mm}^2$$

$$\Delta l = \frac{\sigma l}{E} = 1,587 \text{ mm} \quad , f = \frac{\Delta l}{\sin 45^\circ} = 2,2448 \text{ mm}$$



$$I = \frac{bh^3}{12} = \frac{1,5 \cdot 5^3}{12} = 15,625 \text{ cm}^4$$

$$W = \frac{I}{c} = 6,25 \text{ cm}^3$$

$$\sigma = \frac{M}{W} = \frac{6}{6,25} = 0,96 \text{ kN/cm}$$

③

$$I_{\min} = \frac{20 \cdot 2,5^3}{12} = 26,04 \text{ cm}^4$$

$$A = 20 \cdot 2,5 = 50 \text{ cm}^2$$

$$i = \sqrt{\frac{I}{A}} = 0,722 \text{ cm}$$

$$\lambda = \frac{2 \cdot 350}{0,722} = 969$$

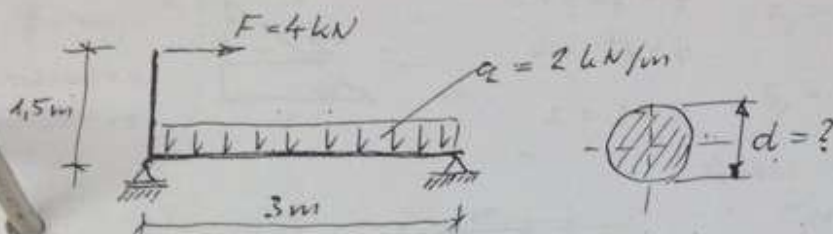
Euler:

$$F_k = \left(\frac{\pi}{\lambda} \right)^2 E I_{\min} = \frac{\pi^2}{700^2} 1000 \cdot 26,04$$

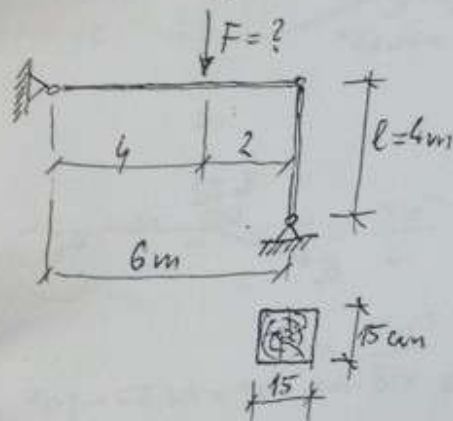
$$\underline{F_k = 0,524 \text{ kN}}$$

4. test iz Mehanike - gozdarji 26.99

- 1) Dimenzionirajte nosilec na upogib, če je dopustna napetost $1,2 \text{ kN/cm}^2$! Prerez nosilca je krog.



- 2) Izračunajte maksimalno silo, da se stebel še ne ukloni! Steber je lesen, kvadratnega prereza.

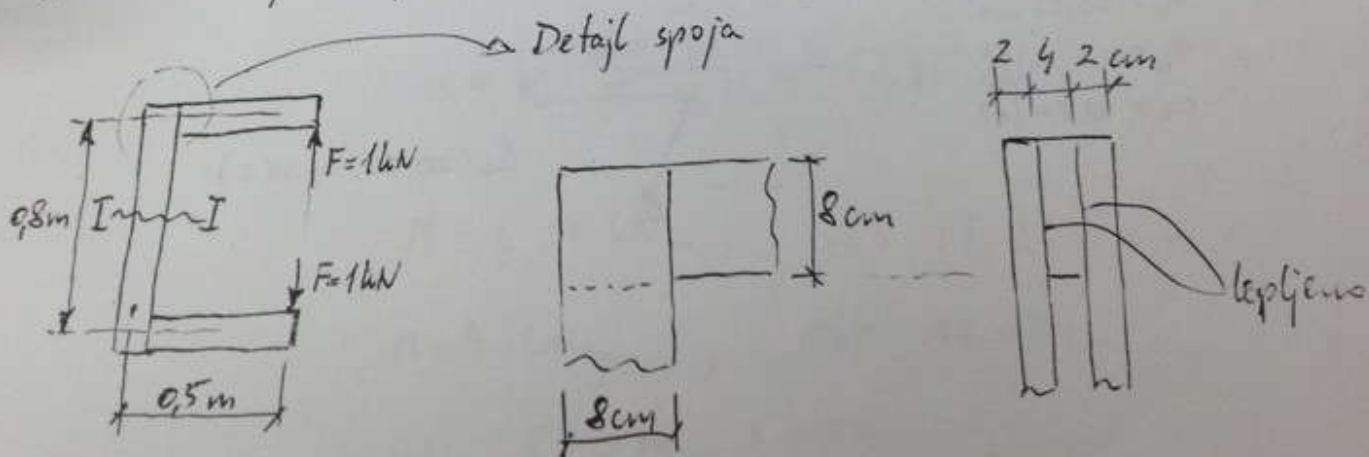


$$E = 1500 \text{ kN/cm}^2$$

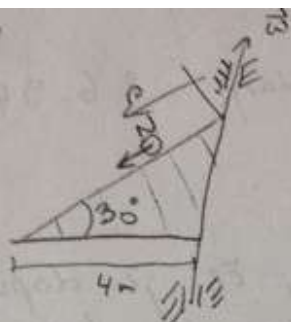
$$\tau_{\text{dop}} = 1,5 \text{ kN/cm}^2$$

$$\nu = 3$$

- 3) Kontrolirajte napetosti v spojih in v označenem prerezu (I-I)



3) 55%



relativno

$$s = \frac{1}{2} t^3 = 4$$

$$\dot{s} = \frac{3}{2} t^2 = 6$$

$$\ddot{s} = 3t = 6$$

Uočeno je da je po zadanu

$$s(t) = \frac{1}{2} t^3 \text{ [m]}$$

$$\varphi(t) = t^2 + 1 \text{ [rad]}$$

Izračunaj: v_a , a_a tačke N po 2. z. g.

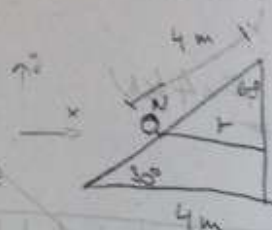
bristeno

$$\varphi = t^2 + 1 = 5$$

$$\dot{\varphi} = 2t = 4$$

$$\ddot{\varphi} = 2 = 2$$

po 2. z. g.



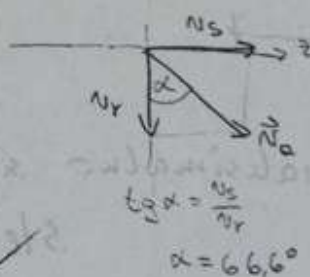
$$v_r = 6 \text{ m/s}$$

$$v_s = \dot{\varphi} \cdot r = 4 \cdot 4 \sin 60^\circ = 13,86 \text{ m/s}$$

$$\vec{v}_a = \vec{v}_r + \vec{v}_s$$

$$\vec{v}_r \perp \vec{v}_s$$

$$v_a = \sqrt{v_r^2 + v_s^2} = 15,1 \text{ m/s}$$



$$a_c = \dot{\varphi}^2 \cdot r$$

$$a_n = \dot{\varphi}^2 \cdot r = \frac{v_s^2}{r}$$

$$Q_r = 6 \text{ m/s}^2$$

$$Q_s^n = \dot{\varphi}^2 \cdot r = 16 \cdot 4 \cdot \sin 60^\circ = 55,43 \text{ m/s}^2$$

$$Q_s^t = \ddot{\varphi} \cdot r = 2 \cdot 4 \cdot \sin 60^\circ = 6,93 \text{ m/s}^2$$

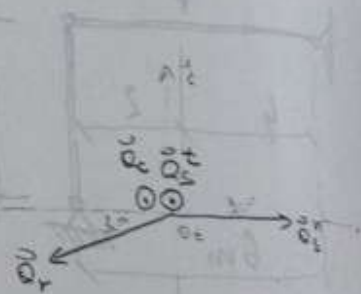
$$Q_c = 2 \omega_s \cdot v_r \cdot \sin \gamma = 2 \cdot \dot{\varphi} \cdot v_r \cdot \sin 120^\circ = 2 \cdot 4 \cdot 6 \cdot \sin 120^\circ = 41,57 \text{ m/s}^2$$

$$Q_a = \sqrt{3^2 + 50,23^2 + 48,5^2} = 69,89 \text{ m/s}^2$$

$$a_x = 50,23 \text{ m/s}^2$$

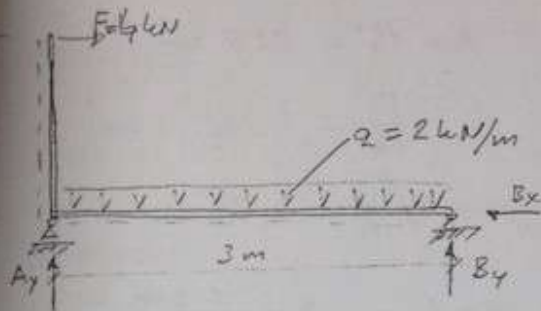
$$a_y = -3 \text{ m/s}^2$$

$$a_z = 48,5 \text{ m/s}^2$$



$$\vec{a}_a(50,23, -3, 48,5)$$

$$\sigma_{\text{day}} = 1,2 \text{ kN/cm}^2$$



Realceipe :

$$B_x = 4 \text{ kN}$$

$$-F \cdot 1,5 - A_y \cdot 3 + 2 \cdot 3 \cdot 1,5 = 0$$

$$A_y = \frac{2 \cdot 3 \cdot 1,5 - 4 \cdot 1,5}{3} = 1 \text{ kN}$$

$$B_y = 2 \cdot 3 - A_y = 5 \text{ kN}$$

Prereez: $0 \leq x \leq 3$



$$M + 2 \cdot \frac{x^2}{2} - 5x = 0$$

$$M = -x^2 + 5x \quad \begin{matrix} M(0) = 0 \\ M(3) = 6 \end{matrix}$$

$$\frac{dM}{dx} = -2x + 5 = 0$$

$$2x = 5 ; \quad x = \frac{5}{2} = 2,5 \text{ m}$$

$$M(2,5) = 6,25 \text{ kNm}$$



$$J = \frac{\pi d^4}{64}$$

$$W = \frac{\pi d^3}{32}$$

$$\sigma = \frac{M}{W} = \frac{6,25 \cdot 32 \cdot 100}{\pi d^3} = 1,2$$

$$d = \sqrt[3]{\frac{625 \cdot 32}{\pi \cdot 1,2}} = 17,44 \text{ cm}$$

$$M = 6,00 \text{ kNm}, \quad d = 17,205 \text{ cm}$$

$$B = 4 \text{ kNm}, \quad d = 15,03 \text{ cm}$$

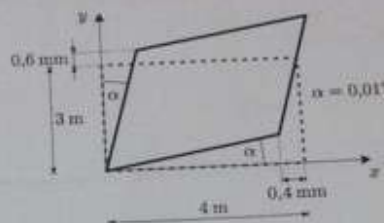
$$W = 520,76 \text{ cm}^3$$

7. IZPIT IZ TRDNOSTI
NTF Geotehnologija in rudarstvo VSS
3. september 2014

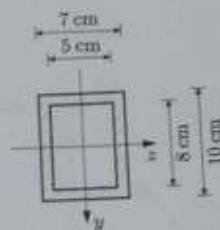
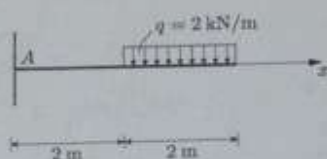
Naloge so enakovredne. Ta list je vaš. Skice, ki definirajo nastopajoče veličine, sodijo v vaš izdelek. V le-tem naj bo razviden postopek reševanja.

- (1) Stena se deformira v ravnini xy , kakor prikazuje slika, in pojavi se ravninsko deformacijsko stanje.

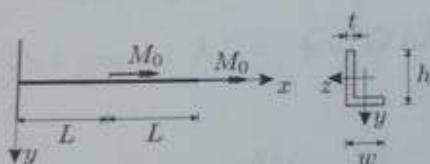
- (a) Zapišite tenzor deformacij.
(b) Izračunajte glavni normalni deformaciji.
(c) Določite smeri glavnih normalnih deformacij, kot kota ki ju smeri oklepata z osjo x .



- (2) Za konzolni nosilec na sliki levo s prerezom na sliki desno izračunajte reakcije v vpetju, vztrajnostni moment prereza in največjo upogibno napetost v konstrukciji.



(3)

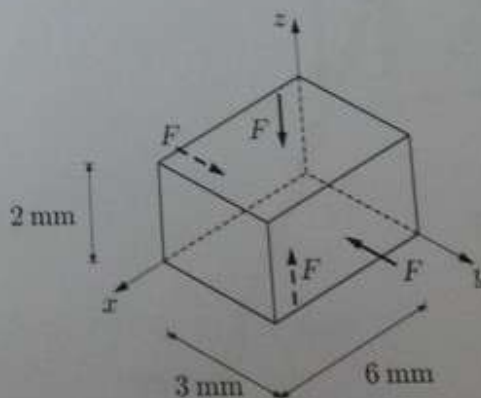


Konzolno vpeti nosilec je obremenjen na sredini in na prostem koncu s torzijskima momentoma velikosti $M_0 = 200 \text{ Nm}$. Dimenzije nosilca so $h = 120 \text{ mm}$, $w = 80 \text{ mm}$, debelina obeh sten je $t = 8 \text{ mm}$, polovica dolžine pa je $L = 0,5 \text{ m}$. Modul elastičnosti gradiva je $E = 210 \text{ GPa}$ in Poissonov količnik je $\nu = 0,3$.

- (a) Izračunajte največjo strižno napetost, ki se v prerezu pojavi zaradi torzije.
(b) Izračunajte zasuk prostega konca.

- (4) Klada je obremenjena po ploskvah z enakomerno porazdeljeno silo $F = 250 \text{ N}$, kakor prikazuje slika.

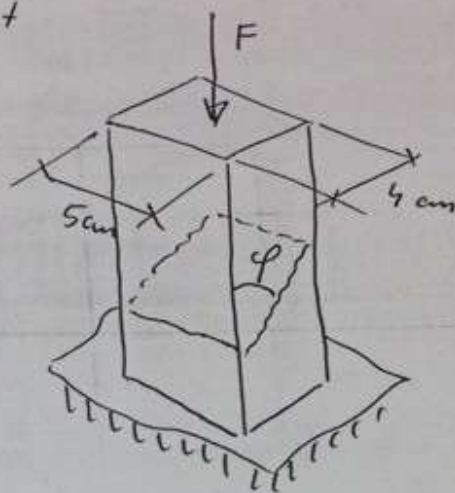
- (a) Določite velikost napetosti, ki ploskve obremenjuje.
(b) Zapišite napetostni tenzor za to napetostno stanje.
(c) Izračunajte spremembo dolžine stranic klade, ki ležijo v smeri osi x .



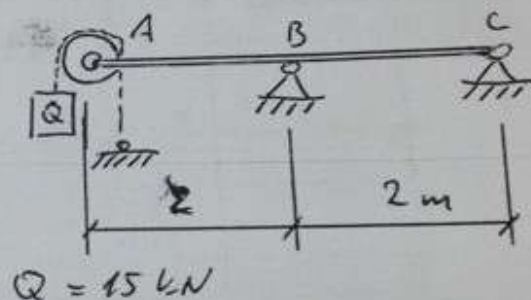
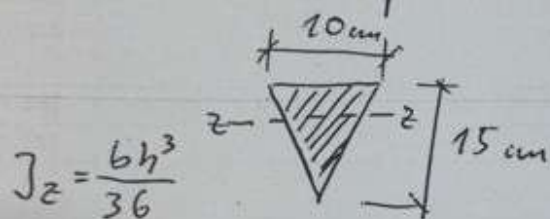
Modul elastičnosti klade je $E = 10 \text{ GPa}$, Poissonovo število pa $0,3$.

Izpit iz Mehanike - 4.1.2007
 OGR - OTM - trdnost

- 1) Izračunajte komponenti napetostnega tenzorja vektorja na ravninah, ki glede na vertikalno obkrepajo kot φ ; ($\varphi = 0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$)!



- 2) Določite mesto in velikost elastičnih normalnih napetosti

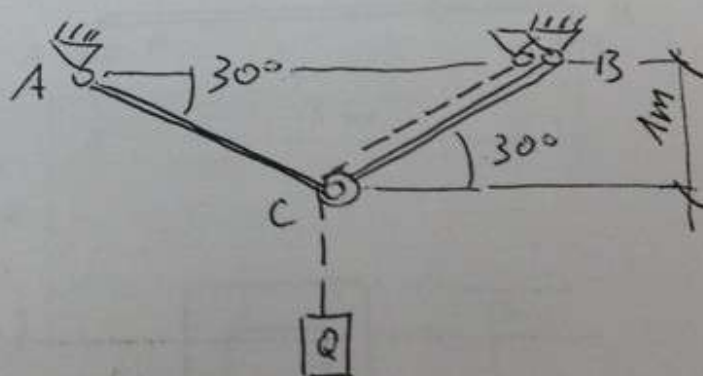


- 3) Izračunajte premik točke C!

$$Q = 100 \text{ kN}$$

$$E = 210000 \text{ MPa}$$

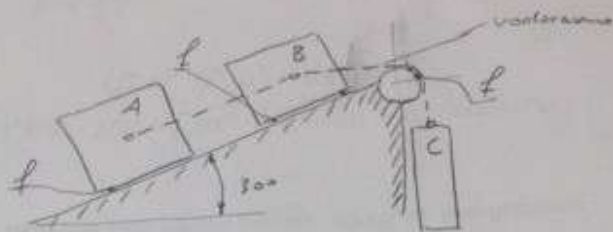
Izmere kolokta v C zanemarite.



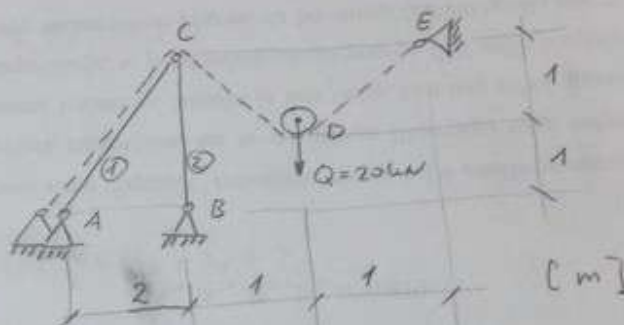
2. kolokvij iz mehanike

lesarji - 11.2.98
 gozdarji - 12.3.98

- 4, 9, 1) Določite maksimalno
 težo telesa C tako,
 da bo sistem v
 ravnovesju! $A=B=300\text{N}$
 $f = 0,4$



- 4, 2) Določite premike
 točke C (x_C, y_C)!
 $A_1 = A_2 = 3\text{cm}^2$
 $E = 21000\text{ kN/cm}^2$

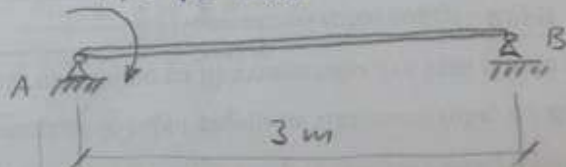


- 4, 3) Določite zasle
 nosilca v točki
 A!

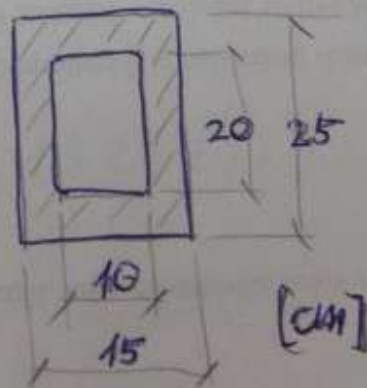
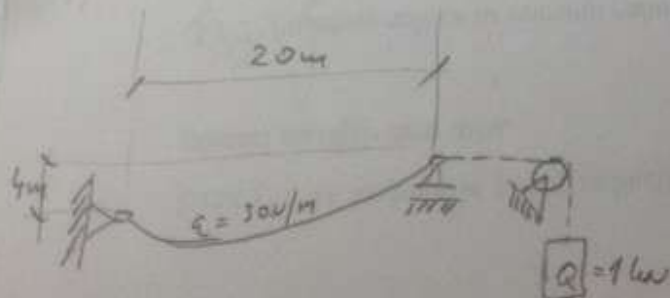
$E = 1000\text{ kN/cm}^2$

uporab!

$M = 12\text{ kNm}$

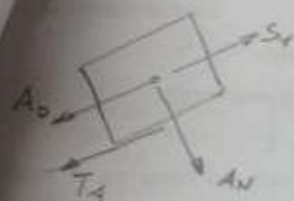


- 4, 3) Narišite sistem se nahaja v
 ravnovesju. Izračunajte ligo
 frena vrvi dolžino vrvi, maks.
 silo v vrvi. (Parabolična oblika)



$$A = B = 300 \text{ N}$$

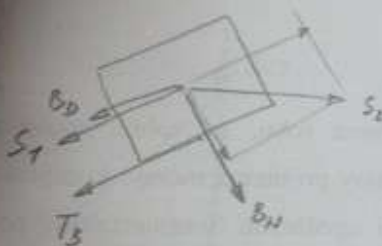
$$\mu = 0,4$$



$$S_1 = A_D + T_A = A \cdot \sin 30^\circ + \mu \cdot A \cdot \cos 30^\circ$$

$$S_1 = 253,923 \text{ N}$$

B:



$$S_2 \cdot \cos \alpha = B_D + S_1 + T_B$$

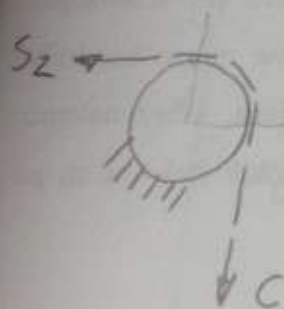
$$T_B = \mu N_B = \mu (B_N + S_2 \cdot \sin \alpha)$$

$$T_B = \mu (B \cdot \cos \alpha + S_2 \cdot \sin \alpha)$$

$$S_2 \cos \alpha = B \cdot \sin \alpha + S_1 + \mu B \cos \alpha + \mu S_2 \sin \alpha$$

$$S_2 = \frac{B \sin \alpha + S_1 + \mu B \cos \alpha}{(\cos \alpha - \mu \sin \alpha)}$$

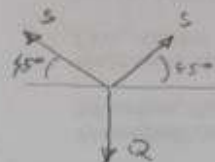
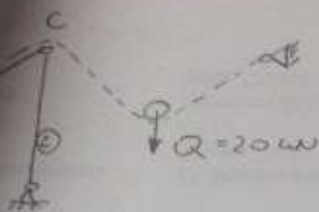
$$S_2 = 762,5 \text{ N}$$



$$C > S_2 \quad ; \quad C = S_2 \cdot e^{\mu \alpha} = 762,5 \cdot e^{0,4 \cdot \frac{\pi}{2}}$$

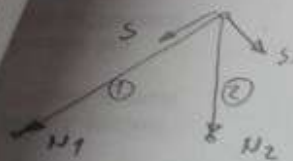
$$C = 1429,277 \text{ N}$$

Vsi koti so 45°



$$Q = 2 \cdot S \cdot \sin 45^\circ$$

$$S = 14,142 \text{ kN}$$

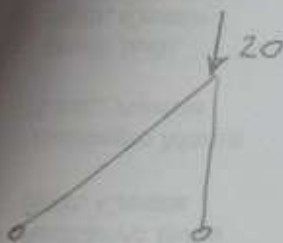


$$S \cos 45^\circ - S \cos 45^\circ - N_1 \cos 45^\circ = 0$$

$$N_1 = 0$$

$$-S \sin 45^\circ - S \sin 45^\circ - N_2 = 0$$

$$N_2 = 20 \text{ kN}$$

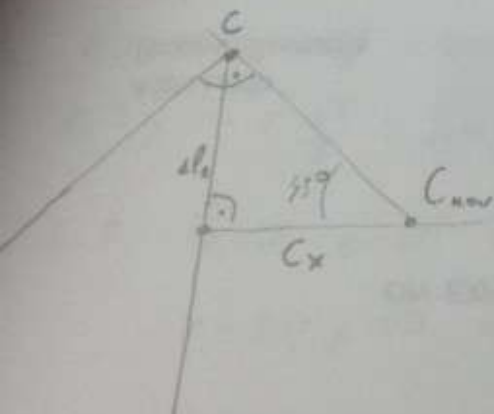


$$\Delta l_1 = 0$$

$$\Delta l_2 = l_2 \cdot \frac{N}{AE} = 200 \cdot \frac{20}{3 \cdot 21000}$$

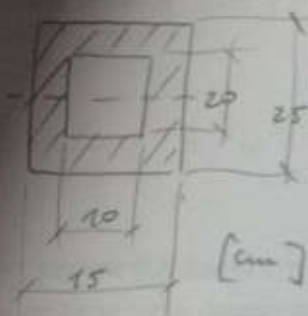
$$\Delta l_2 = 0,06369 \text{ cm}$$

$$\frac{N}{A} = E \frac{\Delta l}{l}$$



$$C_y = -\Delta l_2 = -0,06369 \text{ cm}$$

$$C_x = \Delta l_2 = 0,06369 \text{ cm}$$



$$J = \frac{15 \cdot 25^3}{12} - \frac{10 \cdot 20^3}{12} = 12864,6 \text{ cm}^4$$

$$W = \frac{J}{c} = 1029,17 \text{ cm}^3$$

$$\tau_{\max} = \frac{1}{2} \frac{M}{W} = \frac{1200}{1029,17}$$

$$\sigma_{\max} = \pm 1,166 \text{ kN/cm}^2$$

$$y'' = \frac{-M(x)}{EJ}$$

$$EJ y'' = 4x - 1200$$

$$EJ y' = \frac{1}{2} x^2 - 1200x + C_1$$

$$EJ y = \frac{2}{3} x^3 - 1200 \frac{x^2}{2} + C_1 x + C_2$$

$$\text{R.P.: } x=0, y=0 \Rightarrow C_2=0$$

$$x=300, y=0 \Rightarrow C_1 = 120000$$

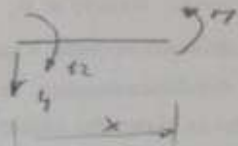
$$y' = \frac{1}{EJ} \left[\frac{1}{2} x^2 - 1200x + 120000 \right]$$

$$x=0; y' = 9,328 \cdot 10^{-3} \text{ rad}$$

$$C = 23 - 3 \cdot 0 = 23$$

$$B = 4 \text{ kN}$$

$$A = 0$$



$$17 - 12 + 4x = 0$$

$$M(x) = -4x + 1200 \text{ [kNcm]}$$

Zasutí v točce A:
0,5344°

površ : - tangenta je vodoravna

$$y' = \frac{1}{EI} [2x^2 - 1200x + 120000]$$

$$2x^2 - 1200x + 120000 = 0$$

$$x^2 - 600x + 60000 = 0$$

$$x_{1,2} = \frac{600 \pm 346,4}{2}$$

$$x_1 = 473 \text{ cm}$$

$$x_2 = 126,8 \text{ cm}$$

iznos povrsa :

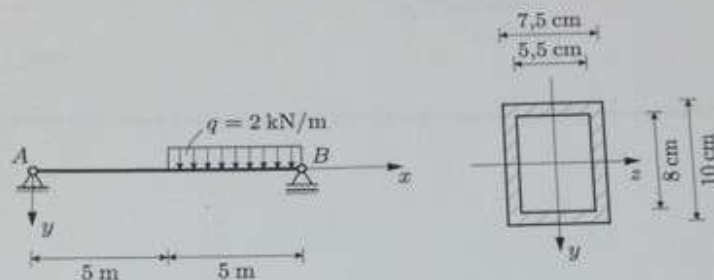
$$y = \frac{1}{EI} \left[\frac{2}{3} x^3 - 600x^2 + 120000x \right]$$

$$y(x = 126,8 \text{ cm}) = \frac{6927630}{1000 \cdot 12864,6} = 0,538 \text{ cm}$$

1. IZPIT IZ TRDNOSTI
NTF Geotehnologija in rudarstvo UN
30. januar 2012

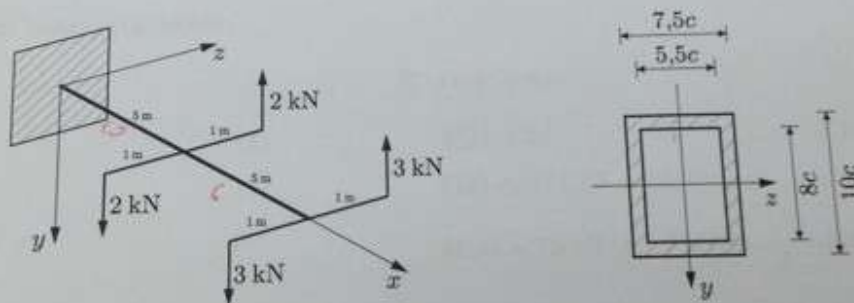
Čas pisanja je 90 min. Naloge so vredne: prva 35, druga 30 in tretja 35 točk. Tega lista ne oddajate. Skice, ki definirajo nastopajoče veličine, sodijo v vaš izdelek. V le-tem naj bo razviden postopek reševanja.

- (1) Določite največjo upogibno napetost v narisanim nosilcu. Upoštevajte, da je nosilec skatlastega profila, prikazanega na sliki desno.



- (2) Za konzolno vpeti nosilec:

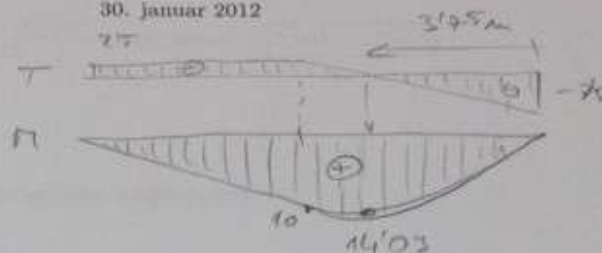
- (a) narišite diagram notranjega torzijskega momenta vzdolž osi x .
(b) Določite velikost parametra c profila na sliki desno, da torzijska napetost nikjer ne bo presegala $\tau_{\text{dop}} = 115 \text{ MPa}$.



1. IZPIT IZ TRDNOSTI – REŠITVE
NTF Geotehnologija in rudarstvo VSS
30. januar 2012

1 Upogib

$$\sigma_{\text{up}}^{\text{max}} = \frac{M^{\text{max}}}{I_z} y^{\text{max}}$$



1.1 Določitev M^{max}

Določimo reakcije, izrišemo diagrama notranje prečne sile in notranjega upogibnega momenta in najdemo ekstrem momenta.

1.1.1 Reakcije

$$A_x = 0 \quad (2)$$

$$A_x = 0 \quad (5)$$

$$A_y + B_y - 2 \text{ kN/m} \cdot 5 \text{ m} = 0 \quad (3)$$

$$A_y = 2,5 \text{ kN} \quad (6)$$

$$-7,5 \cdot 2 \text{ kN/m} \cdot 5 \text{ m} + B_y \cdot 10 = 0 \quad (4)$$

$$B_y = 7,5 \text{ kN} \quad (7)$$

1.1.2 Diagrami notranjih veličin

$$\text{I. } s \in [0, 5 \text{ m}]$$

$$\text{II. } s \in [0, 5 \text{ m}]$$

$$N(s) = 0 \quad (8)$$

$$N(s) = 0 \quad (11)$$

$$T(s) = 2,5 \text{ kN} \quad (9)$$

$$T(s) = -7,5 \text{ kN} + 2 \text{ kN/m} \cdot s \quad (12)$$

$$M(s) = 2,5 \text{ kN} \cdot s \quad (10)$$

$$M(s) = 7,5 \text{ kN} \cdot s - 2 \text{ kN/m} \cdot \frac{s^2}{2} \quad (13)$$

1.1.3 Določimo ekstrem momenta

$$T(s_0) = 0 \Rightarrow M(s_0) = M^{\text{max}} \quad (14)$$

$$T_{\text{II}}(s_0) = -7,5 \text{ kN} + 2 \text{ kN/m} \cdot s_0 \Rightarrow s_0 = \frac{7,5 \text{ kN}}{2 \text{ kN/m}} = 3,75 \text{ m} \quad (15)$$

$$M(s_0) = 7,5 \text{ kN} \cdot 3,75 \text{ m} - 2 \text{ kN/m} \cdot \frac{(3,75 \text{ m})^2}{2} = 14,06 \text{ kNm} = M^{\text{max}} \quad (16)$$

1.2 Določitev I_z in $y^{\max}$

$$I_z = \frac{7,5 \text{ cm} \cdot (10 \text{ cm})^3}{12} - \frac{5,5 \text{ cm} \cdot (8 \text{ cm})^3}{12} = 390,33 \text{ cm}^4 = \boxed{3,903 \cdot 10^{-6} \text{ m}^4} \quad (17)$$

$$y^{\max} = 5 \text{ cm} = \boxed{0,05 \text{ m}} \quad (18)$$

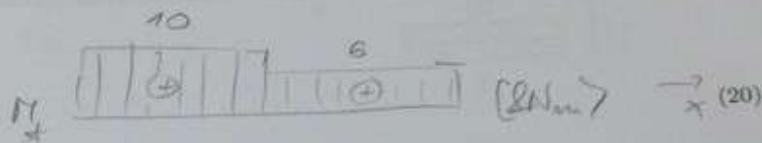
1.3 Določitev največje upogibne napetosti

Vstavimo znano v enačbo (1).

$$\sigma_{\text{up}}^{\max} = \frac{14,09 \text{ kNm}}{3,903 \cdot 10^{-6} \text{ m}^4} \cdot 0,05 \text{ m} = \boxed{180 \text{ MPa}} \quad (19)$$

2 Torzija

$$\tau_t^{\max} = \frac{M_t^{\max}}{W_t}$$



2.1 Diagram notarnjega torzijskega momenta

Najprej izračunamo reakcijo v vpetju, nato nosilec prerežemo v I. in II. polju.

2.1.1 Reakcija M_A

Na vsaki ročici se pojavi moment $2Fr$ in obakrat v pozitivni smeri.

$$M_A + 2 \cdot 2 \text{ kN} \cdot 1 \text{ m} + 2 \cdot 3 \text{ kN} \cdot 1 \text{ m} = 0 \Rightarrow M_A = \boxed{-10 \text{ kNm}} \quad (21)$$

2.2 Določitev $M^{\max}$

Določimo reakcije, izrišemo diagrama notranje prečne sile in notranjega upogibnega momenta in najdemo ekstrem momenta.

I. $s \in [0, 5 \text{ m}]$ (pri vpetju)

I. $s \in [0, 5 \text{ m}]$

$$\xrightarrow{-10 \text{ kNm}} \xrightarrow{M_t}$$

$$\xleftarrow{M_{t1}} \xrightarrow{6 \text{ kNm}}$$

$$M_t = \boxed{10 \text{ kNm}} \quad (22)$$

$$M_{t1} = \boxed{6 \text{ kNm}} \quad (23)$$

$$M^{\max} = \boxed{10 \text{ kNm}} \quad (24)$$

2.3 Določitev W_t

$$W_t = (A_{\text{zun}} + A_{\text{not}}) t_{\min} = (10c \cdot 7,5c + 8c \cdot 5,5c)c = \boxed{119c^3} \quad (25)$$

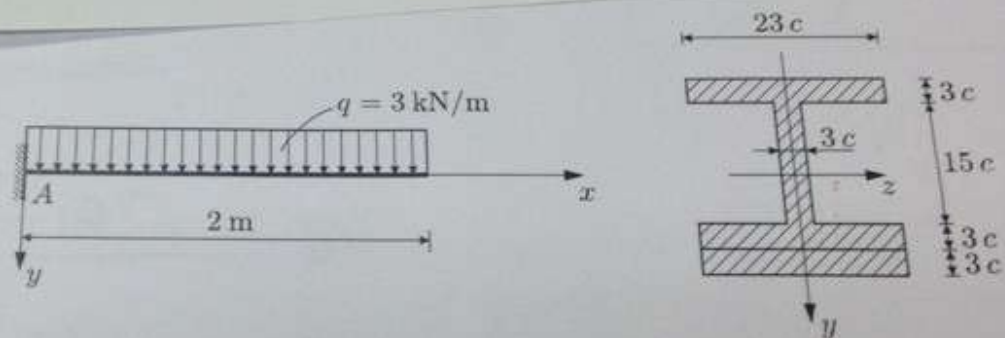
2.4 Dimenzioniranje

$$\tau_t^{\max} = \tau_{\text{dop}}$$

$$\frac{M_t^{\max}}{W_t} = \tau_{\text{dop}}$$

$$\frac{10 \text{ kNm}}{119 c^3} = 115 \text{ MPa}$$

$$c = \sqrt[3]{\frac{10 \text{ kNm}}{119 \cdot 115 \text{ MPa}}} = \boxed{9,00 \text{ mm}}$$

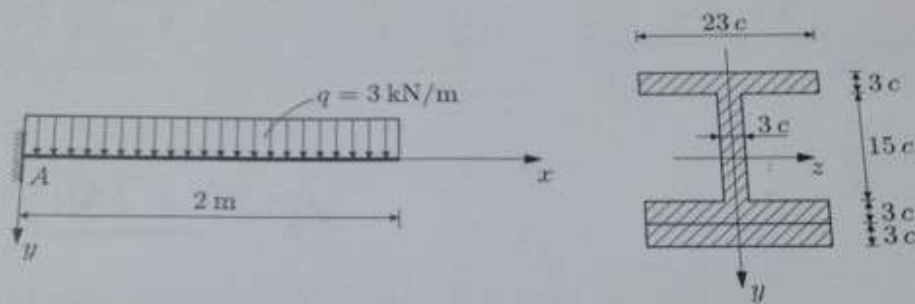


2. IZPIT IZ TRDNOSTI

13. februar 2012

Naloge so vredne: prva 35, druga 35 in tretja 30 točk. Tega lista ne oddajate. Skice, ki definirajo nastopajoče veličine, sodijo v vaš izdelek. V le-tem naj bo razviden postopek reševanja.

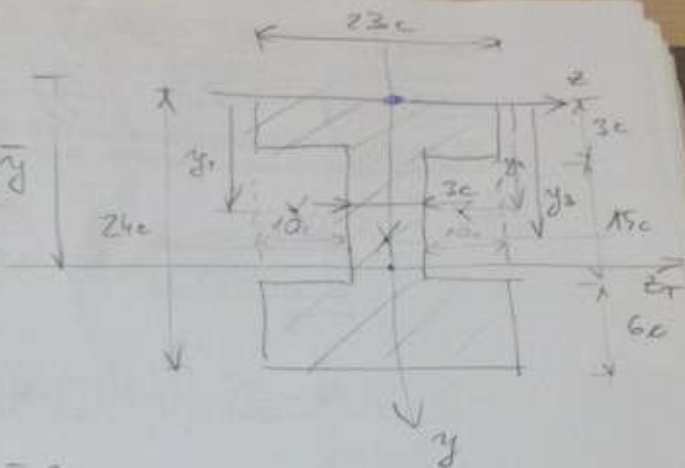
- (1) Konzolni nosilec je grajen iz I-profila, ki mu je v ojačitev privarjena dodatna pasnica (slika desno). Dimenzionirajte konstrukcijo (določite c), če je največja dopustna napetost tega materiala $\sigma_{dop.} = 150 \text{ MPa}$. Vpliv prečne sile zanemarite. Namig: prerez je mogoče obravnavati kot sestav treh pravokotnikov.



$$I_{2T}(c) = ?$$

ES: SCE

y_i	A_i	$y_i A_i$	$\bar{y}$
$105c$	$-150c^2$	$-1575c^3$	
$105c$	$-150c^2$	$-1575c^3$	
$12c$	$552c^2$	$6624c^3$	
	<u>$252c^2$</u>	<u>$3474c^3$</u>	



$$y_1 = 3c + \frac{15c}{2} = 105c = y_1$$

$$y_2 = \frac{24c}{2} = 12c$$

$$A_1 = A_2 = 10c \cdot 15c = 150c^2$$

$$A_3 = 23c \cdot 24c = 552c^2$$

$$\bar{y} = \frac{3474c^3}{252c^2} = 13.79c$$

VERT. MOM.

$$I_{2T}(c) = I_2(c) - \bar{y}^2 A$$

$$\begin{aligned} I_2(c) &= -2 \cdot I_1 + I_3 = -2 \left(\frac{10c(15c)^3}{12} + (105c)^2 \cdot 150c^2 \right) \\ &\quad + \frac{23c(24c)^3}{12} + (12c)^2 \cdot 552c^2 \\ &= 67284c^4 \end{aligned}$$

$$I_{2T}(c) = 67284c^4 - (13.79c)^2 \cdot 252c^2 = \underline{\underline{19363c^4}}$$

točkah z
mom (20 100).

$$y_{max}(c) = ?$$

$$y_{max}(c) = \bar{y} = 13'79 c$$

$$\frac{6 kNm}{19363 c^4} \cdot 13'79 c = 150 MPa$$

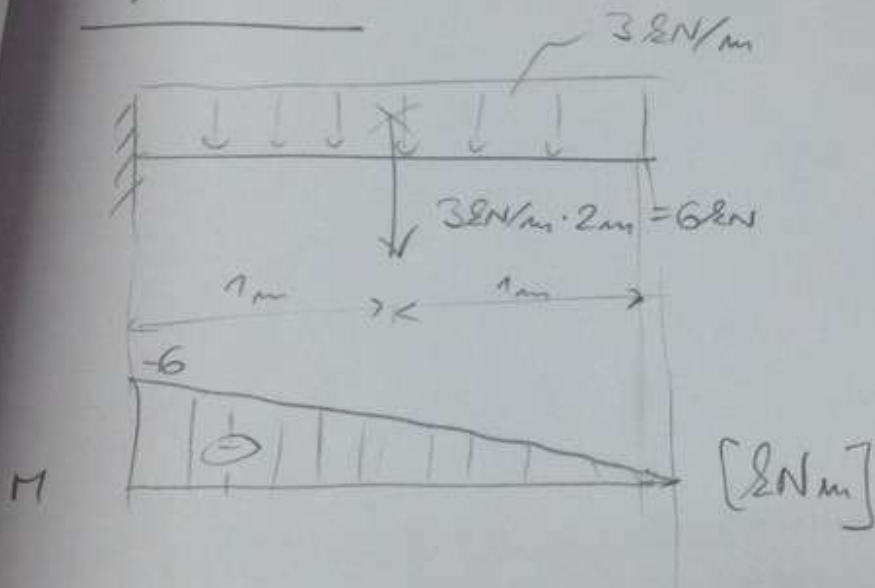
$$c = \sqrt[3]{\frac{6 \cdot 10^3 N \cdot m \cdot 13'79}{19363 \cdot 150 \cdot 10^6 \frac{N}{m^2}}} = \frac{0'00305 m}{1} = \underline{\underline{3'05 mm}}$$

2. 12PIT 12
TRDNOST/

$$\sigma_{\text{up}}^{\text{max}}(c) = \sigma_{\text{dgr}}$$

$$\frac{M^{\text{max}}}{I_{zz}(c)} y^{\text{max}}(c) = \sigma_{\text{dgr}} = 150 \text{ MPa}$$

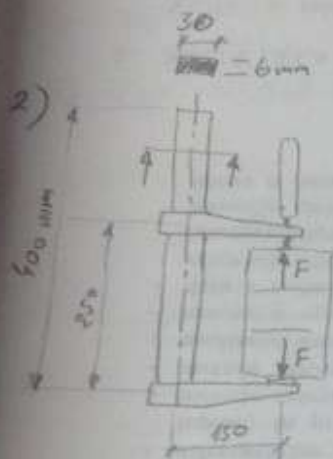
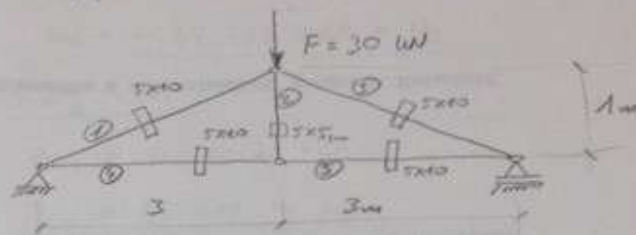
$$M^{\text{max}} = ?$$



$$M^{\text{max}} = |-6 \text{ kNm}| = 6 \text{ kNm}$$

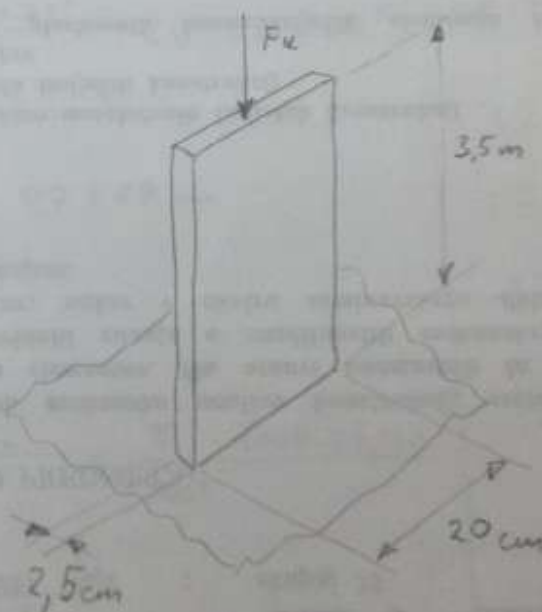
2. test iz Mehanike - lesarji
12.6.35 ??

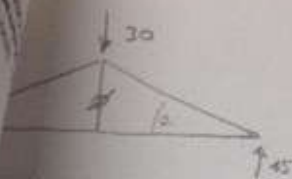
- 1) Izračunajte velikeosti
napetosti v vseh palicah!
 $F = 30 \text{ kN}$



- 2) V naprženem elementu narisane
svore
sprave določite mesto in velikeost
največjih upogibnih napetosti!
 $F = 500 \text{ N}$

- 3) Izračunajte kritično
(ne dopustno) upogibno
silo naprjeno stojee
lesene deske, ki je
vbetonirana v tla!
 $E = 1000 \text{ kN/cm}^2$





$$\frac{4}{3} = \tan \alpha \Rightarrow \alpha = 18,435^\circ$$

$$N_1 \cdot \sin \alpha + 45 = 0$$

$$N_1 = -47,434 \text{ kN} = N_3$$

$$N_2 = 0$$

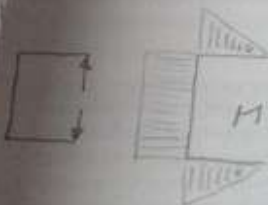
$$N_4 + N_1 \cdot \cos \alpha = 0$$

$$N_4 = 45 \text{ kN} = N_5$$

$$\sigma_2 = \frac{N_1}{A_1} = \frac{-47,434}{50}$$

$$\sigma_2 = -0,948 \text{ kN/cm}^2$$

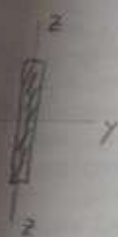
$$\sigma_4 = \sigma_5 = \frac{45}{50} = 0,9 \text{ kN/cm}^2$$



$$M_{\max} = 500 \cdot 150 = 75000 \text{ Nmm}$$

$$W_x = \frac{bh^3 \cdot 2}{12 \cdot h} = \frac{6 \cdot 30^3}{6} = 900 \text{ mm}^3$$

$$\sigma = \frac{75000}{900} = 83,33 \text{ N/mm}^2$$



$$J_{\min} = J_z = \frac{20 \cdot 2,5^3}{12} = 26,0416 \text{ cm}^4$$

$$A = 20 \cdot 2,5 = 50 \text{ cm}^2$$

$$i = \sqrt{\frac{J_{\min}}{A}} = 0,7217 \text{ cm}$$

$$\lambda = \frac{\beta \cdot l}{i} = \frac{2 \cdot 350}{0,7217} \approx 970$$

$$\text{Euler: } F_k = \frac{\pi^2 E J_{\min}}{(\beta \cdot l)^2} = \frac{\pi \cdot 1000 \cdot 26,0416}{700^2}$$

$$F_k = 0,5245 \text{ kN}$$

$$\text{ce } h = 5 \text{ cm}$$

$$J = 260,4 \text{ cm}^4$$

$$i = 1,443 \text{ cm}$$

$$\lambda = 554,4$$

$$F_k = 1,016 \text{ kN}$$

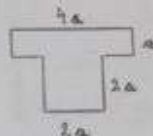
1. IZPIT IZ TRDNOSTI
NTF Geotehnologija in rudarstvo VSS in Gospodarsko geoinženirstvo UN
10. februar 2011

Vse rešitve ustrezno utemeljite. Iz vašega izdelka naj bo razviden postopek reševanja. Naloge so vredne 35+35+30 točk. Profesor in asistent vam želita veliko uspeha.

- (1) Nosilec z dolžino 5 m s prerezom narisane oblike obremenimo s silo $F = 5 \text{ kN}$. Določi izmero a , da upogibne napetosti v nosilcu ne bodo presegle 100 MPa .

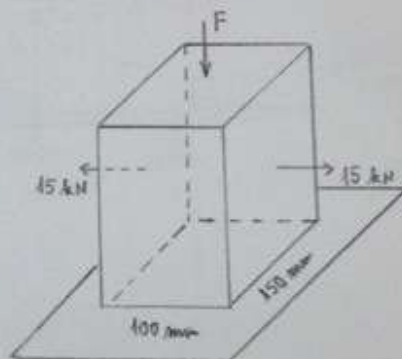


Prerez:



- (2) Na narisani aluminijev kvader dimenzij $100 \text{ mm} \times 150 \text{ mm} \times 200 \text{ mm}$ delujemo na stranskih ploskvah z natezno silo 15 kN , na zgornjo pa s tlačno silo F .

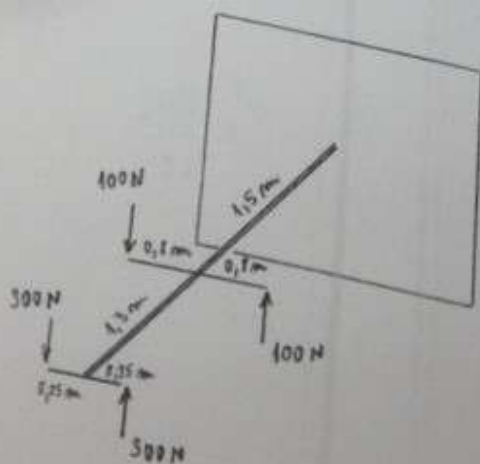
- Kolikšna mora biti sila F , če želimo zmanjšati višino kvadra za $0,002 \text{ mm}$?
- Zapiši napetostni in deformacijski tenzor.
- Koliko dolgi so robovi kvadra pa obremenitvi?



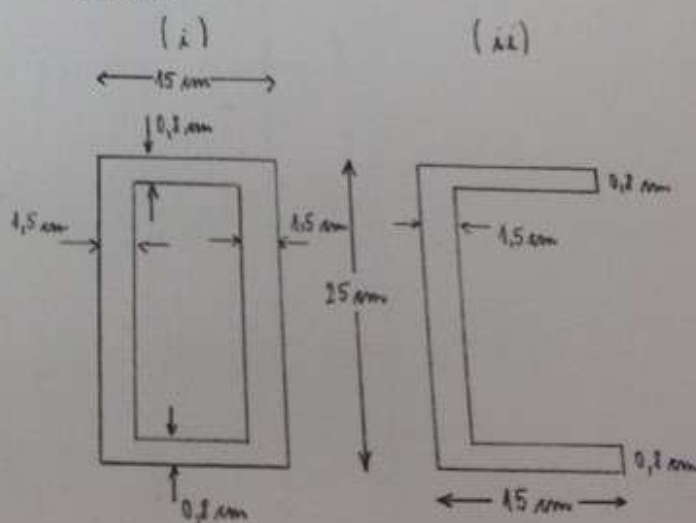
Youngov modul je 70 GPa , Poissonovo število pa $0,3$.

- (3) Vpeto palico obremenimo prek vzvodov z dvojicama sil.

- Nariši diagram torzijskega momenta v gredi.
- Kolikšna je največja torzijska napetost v palici, če je prerez palice tak, kot je narisano na skici.

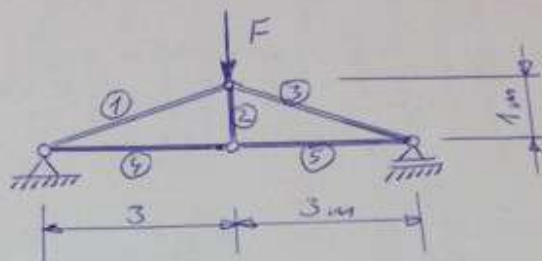


Prereza:

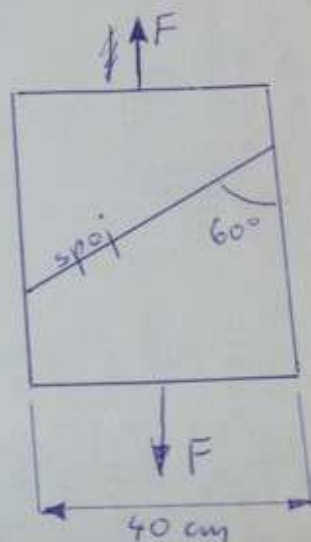


Izpit iz Mehanike - Trdnost, OGR, OTM
31. 1. 2005

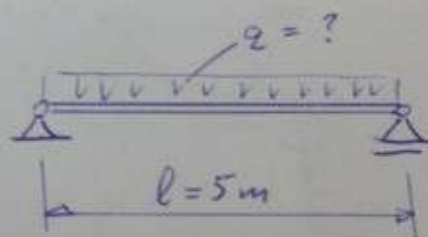
- 1) Izračunajte velikosti napetosti v vseh palicah! $F = 30 \text{ kN}$
Prerez palice je pravokotnik $5 \times 10 \text{ cm}$.



- 2) Plošča debeline 3 cm je zlepljena iz dveh trapeznih plošč. Izračunajte komponente napetostnega vektora na spojni ravnini, če predpostavimo, da vlečna sila $F = 24 \text{ kN}$ deluje enakomerno porazdeljena po obeh ploskvah.



- 3) Kamnita preklada je obremenjena samo z lastno težo. Izračunajte, ali napetost v prekladi presega dopustno, in če, na kolikšnem območju



$$\rho = 2200 \text{ kg/m}^3$$

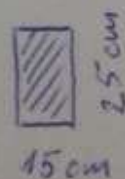
$$E = 5,3 \cdot 10^4 \text{ MPa}$$

$$\nu = 0,25$$

$$\sigma_{\text{dop nateg}} = 1,2 \text{ MPa}$$

$$\sigma_{\text{dop tlak}} = -120 \text{ MPa}$$

prerez preklade:



1)



$$\tan \alpha = \frac{1}{3}; \alpha = 18,435^\circ$$

$$N_1 \cdot \sin \alpha + 15 = 0$$

$$N_1 = -47,434 \text{ kN} = N_3$$

$$N_2 = 0$$

$$N_4 + N_1 \cos \alpha = 0$$

$$N_4 = 45 \text{ kN} = N_5$$

$$\sigma_1 = \sigma_3 = \frac{N_1}{A_1} = \frac{-47,434}{50} = -0,948 \text{ kN/cm}^2$$

$$\sigma_2 = \sigma_4 = \frac{N_5}{50} = 0,9 \text{ kN/cm}^2$$

2)



$$\sigma_x = \frac{84}{3 \cdot 4} = 0,2 \text{ kN/cm}^2$$



$$p = \frac{F \cdot \cos 30^\circ}{A} = 0,1732 \text{ kN/cm}^2$$

$$\sigma_n = p \cdot \cos 30^\circ = 0,15 \text{ kN/cm}^2$$

$$\tau_n = 10,0866 \text{ kN/cm}^2 \text{ (shear)}$$

2. naodm)

$$\sigma_{ij} = \begin{pmatrix} 0,2 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\sigma_n = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos(2\varphi) + \tau_{xy} \sin(2\varphi)$$

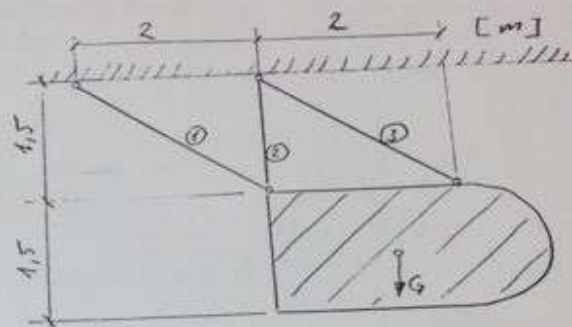
$$\sigma_n = 0,1 + 0,1 \cos(60^\circ) = 0,15 \text{ kN/cm}^2$$

$$\tau_n = -\frac{\sigma_x - \sigma_y}{2} \sin(2\varphi) + \tau_{xy} \cos(2\varphi)$$

$$\tau_n = -0,1 \sin(60^\circ) = -0,0866 \text{ kN/cm}^2 \text{ shear!}$$

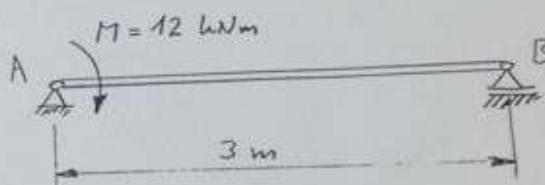
Izpit iz Mehanike
Lesarstvo, VSŠ, 5.2.2003

- 1) Določite sile v palicah 1, 2 in 3, ki nosijo togo homogeno ploščo teže $G=20 \text{ kN}$!

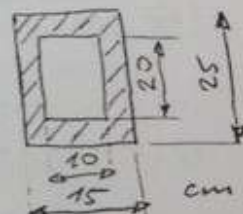


- 2) Določite:

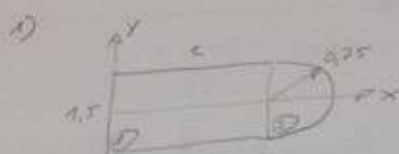
- a) mesto in velikost največjih upogibnih napetosti,
in velikost osi!



prerez nosilca

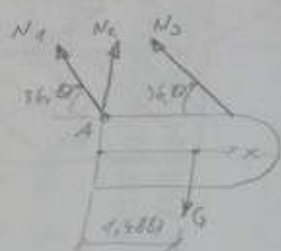


- b) mesto in velikost največjih strižnih napetosti



i	A _i	y _i	A _i · y _i
1	3	1	3
2	9,88	8,75	2,048
Σ	3,88		5,048

$$x_T = \frac{5,048}{3,88} = 1,3 \text{ m}$$



$$\Sigma x: -N_1 \cdot \cos \alpha - N_3 \cdot \cos \alpha = 0$$

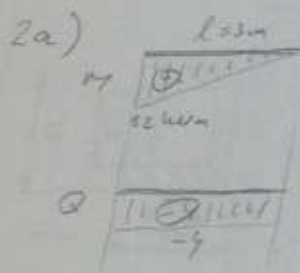
$$\Sigma y: N_1 \cdot \sin \alpha + N_2 + N_3 \cdot \sin \alpha - G = 0$$

$$\Sigma A: -G \cdot x_T + N_3 \cdot \sin \alpha \cdot 2 = 0$$

$$N_3 = \frac{G \cdot x_T}{2 \cdot \sin \alpha} = 21,67 \text{ kN}$$

$$N_1 = -N_3 = -21,67 \text{ kN}$$

$$N_2 = G - N_3 \cdot \sin \alpha - N_1 \cdot \sin \alpha = 20 \text{ kN}$$



$$J_2 = 12865 \text{ cm}^4$$

$$\sigma_{\text{top}} = \pm \frac{12000 \cdot 12,5}{12865} = \pm 1,166 \text{ kN/cm}^2$$

$$S_{y3} = A_3 \cdot y_{T3} = 87,5 \cdot 7,6786 = 671,875 \text{ cm}^3$$



i	A _i	y _i	A _i · y _i
1	5 × 12,5	6,25	390,625
2	10 × 7,5	11,25	281,25
Σ	87,5		671,875

$$y_{T3} = 7,6786 \text{ cm}$$

$$\tau_3 = \frac{Q \cdot S_{y3}}{J_2 \cdot b(z)} = \frac{4 \cdot 671,875}{12865 \cdot 5} = 0,04178 \text{ kN/cm}^2$$

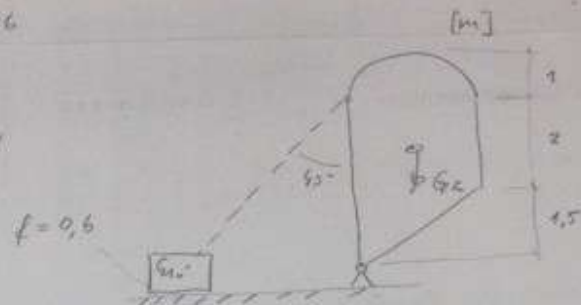
5

izpit iz tehniške mehanike - vsi
10.9.96

1) Izračunajte minimalno
težo telesa G_2 , tako, da
bo sistem v ravnotežju!

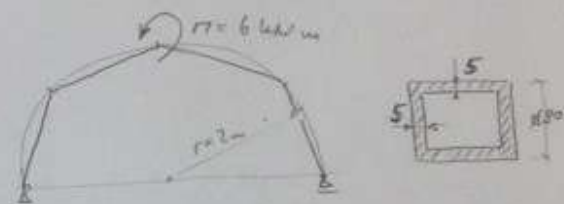
G
T
L

$$G_2 = 30 \text{ kN}$$



2) Narišite diagram upogibnih
momentov ter izračunajte
vrsto in velikost največjega
napetosti, ki so posledica
upogiba

G
T
L

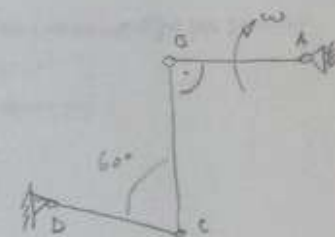


3) Za narisano lego mehanizma
izračunajte kotno hitrost
ročice BC

T
L

$$\omega = 20 \pi \text{ s}^{-1}$$

$$AB = 0.4 \text{ m}, BC = 1.2 \text{ m}, CD = 0.6 \text{ m}$$



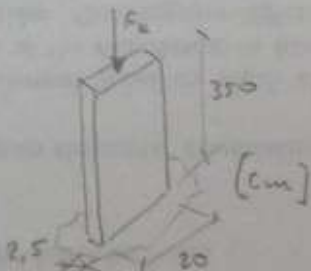
4) Glej izpit - teoretika
iz dne 22.6.95 - Analoga

T



5) Glej test - lesarji
iz dne 12.6.95 - 3. analoga

G

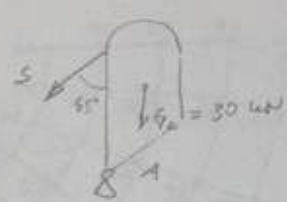




istema coordinato x

i	A_i	x_i	$A_i \cdot x_i$
1	4,72	1	4,72
2	4	1	4
3	4,5	0,66	1
Σ	20,71		6,571

$$x_T = 0,929 \text{ m}$$



$$\Sigma M_A = 0 ; -G_1 \cdot x_T + S \cdot \sin 45^\circ \cdot 3,5 = 0$$

$$S = 11,265 \text{ kN}$$



$$T = S \cdot \cos 45^\circ = N \cdot f = f(G_1 - S \sin 45^\circ)$$

$$N = G_1 - S \cdot \sin 45^\circ$$

$$S(\cos 45^\circ + f \sin 45^\circ) = G_1$$

$$G_1 = 21,24 \text{ kN}$$

2)



$$A - B = 0$$

$$B \cdot 4 = 6$$

$$B = 1,5 \text{ kN}$$

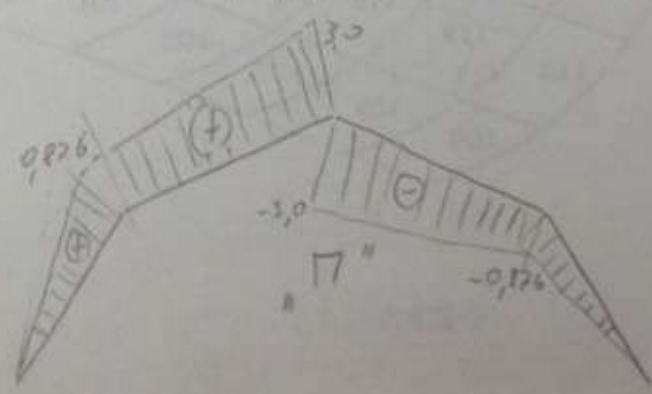


$$x = r - r \cdot \cos 45^\circ = 0,5852 \text{ m}$$

I polare :



$$M(0) = 0, M(0,586) = 0,8786 \text{ kNm}$$



statische Aufgabe



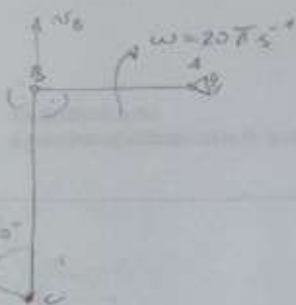
$$J = \frac{1}{12} (80^4 - 60^4) = 144,25 \text{ cm}^4$$

$$W = \frac{J}{y} = 35,3125 \text{ cm}^3$$

$$\sigma = \frac{M}{W} = \frac{300}{35,3125} = 8,495 \text{ kN/cm}^2$$

$$\sigma = 84,95 \text{ MPa}$$

3



$$v_B = \omega \cdot AB = 8\pi \text{ m/s} = 25,12 \text{ m/s}$$

Legen polen $\overline{BP}$ - pol drogu $\overline{BC}$

$$\tan 60^\circ = \frac{\overline{BP}}{\overline{BC}} \Rightarrow \overline{BP} = 1,2 \cdot \tan 60^\circ$$

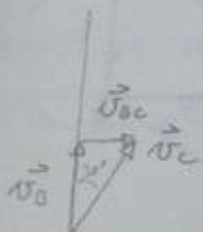
$$\overline{BP} = 2,078 \text{ m}$$

1. uacina : pol

$$\omega_{BC} = \frac{v_B}{\overline{BP}} = \frac{8\pi}{2,078} = 3,879 \pi \text{ s}^{-1} = 12,092 \text{ s}^{-1}$$

2. uacina : diagram

$$\vec{v}_C = \vec{v}_B + \vec{v}_{BC}$$



$$\tan 30^\circ = \frac{v_{BC}}{v_B} \Rightarrow v_{BC} = 8\pi \cdot \tan 30^\circ = 14,51 \text{ m/s}$$

$$\omega_{BC} = \frac{v_{BC}}{\overline{BC}} = 12,092 \text{ s}^{-1}$$

$$v_C = \frac{v_B}{\cos 30^\circ} = 29,0208 \text{ m/s}$$

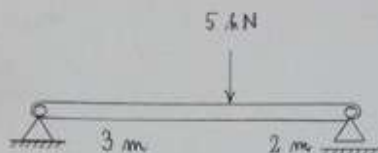
$$\omega_{CO} = \frac{v_C}{\overline{CO}} = 48,37 \text{ s}^{-1}$$

2. IZPIT IZ TRDNOSTI

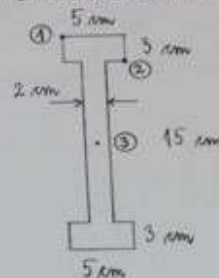
NTF Geotehnologija VSS in Metalurške tehnologije VSP
26. junij 2008

Vse rešitve ustrezno utemeljite. Iz vašega izdelka naj bo razviden postopek reševanja. Naloge so vredne 35+30+35 točk. Profesor in asistent vam želita veliko uspeha.

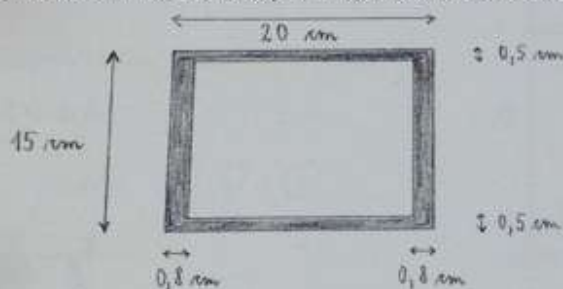
- (1) Nosilec dolžine 5 m s prerezom narisane oblike podpremo in obremenimo s silo 5 kN. Izračunaj, kolikšne upogibne napetosti se pojavijo v točkah ①, ② in ③ v prerezu, ki je 1 m oddaljen od levega roba nosilca.



$I_{MM_{00}}$:



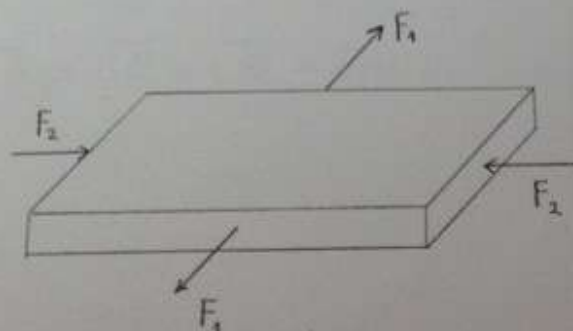
- (2) Izračunaj, s kolikšnim torzijskim momentom sme biti obremenjen nosilec s prerezom skicirane oblike, če želimo, da torzijske napetosti ne presežejo 50 MPa.



- (3) Jekleno ploščo dimenzij 500 mm × 200 mm × 10 mm na stranskih ploskvah obremenimo s silama $F_1 = 20$ kN in $F_2 = 10$ kN v narisanih smereh.

- Če želimo, da se debelina plošče ne spremeni, ali moramo na zgornjo in spodnjo ploskev delovati s kakšno silo? Če je odgovor pritrdilen, določi s kolikšno in v kateri smeri.
- Zapiši tenzorja napetosti in deformacije.
- Izračunaj spremembo prostornine plošče.

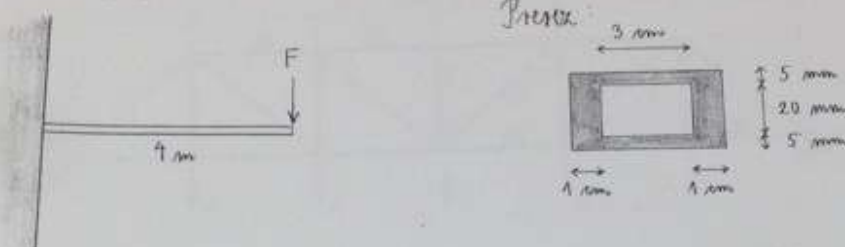
Youngov modul jekla je 200 GPa, Poissonovo število pa 0,3.



1. IZPIT IZ TRDNOSTI
NTF Geotehnologija VSS in Metalurške tehnologije VSP
3. junij 2008

Vse rešitve ustrezno utemeljite. Iz vašega izdelka naj bo razviden postopek reševanja. Naloge so vredne 33+34+33 točk. Profesor in asistent vam želita veliko uspeha.

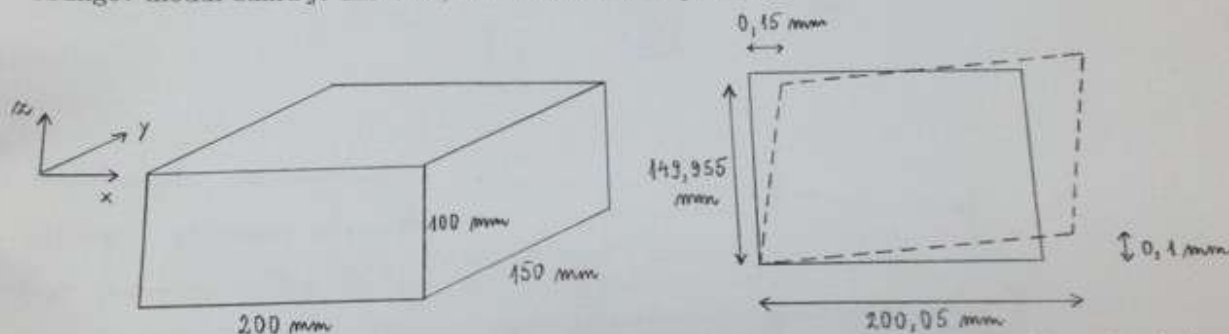
- (1) Nosilec dolžine 4 m s prerezom narisane oblike kozolno vpremo in obremenimo s silo 5 kN. Izračunaj največjo upogibno napetost, ki se pojavi v nosilcu in točno določi mesto, kjer se ta napetost pojavi.



- (2) Bakren kvader dimenzij 200 mm × 150 mm × 100 mm obremenimo tako, da se zgornja (in spodnja) ploskev deformirata na narisani način, medtem ko deformacij v smeri osi z ni.

- (a) Zapiši tenzor deformacije. (V kateri koli točki kvadra je enak.)
(b) Določi glavni normalni deformaciji in njuni smeri (v ravnini xy).
(c) Izračunaj tenzor napetosti.

Youngov modul bakra je 120 GPa, Poissonovo število pa 0,36.



- (3) Vpeta palica z dolžino 2 m, katere prerez je kvadrat s stranico 15 mm, je sestavljena iz dveh delov. Prvi je iz jekla ($E_1 = 200$ GPa, $\alpha_1 = 12 \cdot 10^{-6} \frac{1}{^\circ\text{C}}$) in je dolg 1,25 m, drugi pa je iz medenine ($E_2 = 100$ GPa, $\alpha_2 = 21 \cdot 10^{-6} \frac{1}{^\circ\text{C}}$). To palico segrejemo od 20°C do 50°C.

- (a) S kolikšno tlačno silo moramo delovati na prostem koncu palice, da bo njena dolžina ostala enaka?
(b) Za koliko se pri tem podaljša (skrajša) posamezen del?

